Intelligent System for Indexing Academic Journals Recommended by the Committee (KOKCHBO MHBO PK)

Diploma project: analysis and design report (pre-defense 1, 24 October 2026)

Team: Nurzhan Izimbetov (Backend & Search Core), Abilkhaiyr Sarsenbay (Frontend & Data Integration), group SE-2421. Supervisor: Symbat Nurgaliyeva, PhD. Repository: https://github.com/nurzhqn0/indexing-system

Project definition. The core of the diploma is a web system for searching the Committee's list of recommended journals: hybrid retrieval (full-text, fuzzy and semantic search fused by Reciprocal Rank Fusion), a status check, a versioned registry, a Vue frontend and a FastAPI backend with PostgreSQL. This follows the supervisor's justification.

#	Component	Where
1	Project proposal	Section 1
2	Evidence of relevance	Section 2
3	Literature review	Section 3
4	Existing system comparison	Section 4
5	Technology comparison	Section 5
6	Requirements specification	Section 6
7	Architecture and diagrams	Section 7
8	UX / prototype	Section 8
9	Verification and evaluation plan	Section 9
10	Project management and risks	Section 10
11	Supervisor feedback	Section 11
A-E	Appendices: search log, ADRs, abbreviations, task list, API examples	A, B, C, D, E
-	References (APA 7th edition)	References

Sources of material: the supervisor's justification (roles, plan), the Committee's list file (edition of 15 Sep 2026), `docs/lit/references.bib`, diagrams in `diagrams/`.

1. Project Proposal

Title

EN: Development of an Intelligent System for Indexing Academic Journals Recommended by the Science and Higher Education Quality Assurance Committee of the Ministry of Science and Higher Education of the Republic of Kazakhstan

RU: Разработка интеллектуальной системы индексации научных журналов, рекомендованных Комитетом по обеспечению качества в сфере науки и высшего образования Министерства науки и высшего образования РК

KZ: ҚР Ғылым және жоғары білім министрлігінің сапаны қамтамасыз ету комитеті ұсынған ғылыми журналдарды индекстеуге арналған интеллектуалды жүйесін әзірлеу

Team

Role	Name	Group	Responsibility (from the supervisor's justification)
Student	Nurzhan Izimbetov	SE-2421	**Backend & Search Core Engineer**: server architecture, database, FastAPI, hybrid search, fuzzy search, performance, containerisation and deployment
Student	Abilkhaiyr Sarsenbay	SE-2421	**Frontend & Data Integration Engineer**: Vue frontend, search forms, tables, filters, journal cards, user scenarios, validation, error handling, preparation and synchronisation of registry data
Supervisor	Symbat Nurgaliyeva, PhD, assistant professor		weekly meetings, code review, architecture and search-method consultation

Problem statement

Researchers, teachers and doctoral students in Kazakhstan must regularly check whether a journal is in the list of publications recommended by the Committee (КОКСНВО МНВО РК) for publishing the main results of scientific activity. The list is published as documents that are inconvenient to search, and it changes: it is updated regularly and was reorganised into three levels by order No. 289 of the Minister of Science and Higher Education of 3 June 2025 (section 2). Titles occur in Kazakh, Russian and English, with transliteration variants, abbreviations and typos. A user needs one tool that finds a journal quickly, even from an imperfect query, and shows its status, list level and subject area.

Target users

User	Need
Researchers, teachers, PhD students	check whether a journal is recommended and at which level before submitting
Librarians, dissertation-council staff	verify publication requirements quickly and consistently
Journal editors	see how their journal is listed
Ministry / Committee analysts (secondary)	browse the list by level, subject and language

User needs come from the supervisor's justification and from the literature; they are **assumptions until confirmed** by the user survey and discussions (section 2.4).

Aim

To design, implement, deploy and evaluate a web system that lets users search the Committee's list of recommended journals with hybrid retrieval (full-text, fuzzy and semantic search combined by Reciprocal Rank Fusion) across Kazakh, Russian and English queries, and shows journal status and metadata.

Research question

Does hybrid retrieval, which combines PostgreSQL full-text search, trigram fuzzy matching and multilingual vector search through Reciprocal Rank Fusion, find the intended journal more reliably than each method alone for realistic trilingual queries (exact titles, partial titles, typos, transliteration, subject descriptions), while meeting an interactive latency target?

Hypotheses

Thresholds are fixed before any measurement and are not changed after seeing test results (sections 6.4, 9.4). Metrics and the decision rule are defined in section 9.4.

ID	Hypothesis	Success criterion
HS1	Hybrid retrieval (H1: FTS + trigram + vector, fused by RRF) finds the intended journal more reliably than every single method	MRR@10 of H1 is at least 0.05 above the best of M1-M4 on the whole test set; the paired-bootstrap 95% confidence interval of the difference excludes 0
HS2	Hybrid retrieval stays reliable on imperfect queries	Recall@5 of H1 is at least 0.95 overall and at least 0.90 on typo queries
HS3	Hybrid retrieval is fast enough for interactive use	p95 response time of H1 is at most 300 ms on the deployment machine, warm cache, CPU embedding

HS1 is the riskiest: the registry is small (243 rows), so the best single method may already score close to the maximum. A pilot run in November 2026 checks whether the +0.05 margin is reachable (section 9.7), and the CPU latency of the candidate embedding models is measured at the same time (section 5.3, task DIP-38).

Objectives

#	Objective	Owner
O1	Obtain, normalise and version the registry data (journals, ISSN, publisher, list level, subject, language, inclusion order and date) with a repeatable import	Abilkhaiyr (data), Nurzhan (schema)
O2	Design and implement the backend: schema, FastAPI with documented endpoints, import endpoint	Nurzhan
O3	Implement and tune the search core: FTS, trigram fuzzy search, vector search, RRF fusion	Nurzhan
O4	Implement the Vue frontend: search, filters, tables, journal card, status check, validation, error handling	Abilkhaiyr
O5	Build the evaluation set and measure relevance and speed against single-method baselines	Nurzhan (search side), Abilkhaiyr (query set, UI checks)
O6	Containerise and deploy the system; write documentation and tests	both

Open scope questions

1. Which parts of the list are in scope for search: Lists 1-3 only, or also the list of security and law-enforcement journals, conference proceedings and trusted publishers?
   Current working answer: Lists 1-3 and the military and security list are imported; the evaluation focuses on the IT/CS subset.

2. Should the thousands of international journals that enter List 1 automatically through Web of Science and Scopus be searchable? Current working answer: no, only a rule note (risk R4).

Expected outcomes

- Deployed web service with documented API and relevant, fast search over the registry.
- Versioned registry database and repeatable import and update module.
- Vue interface with search, filtering, journal pages and status check.
- Evaluation report: relevance and latency of hybrid versus baselines, per language and query type.
- Documentation, tests, container setup.

Scope

In scope

- Journals of the Committee's list: the whole file is imported (Lists 1, 2, 3 and the list of military and security university journals, 243 rows in the edition of 15 Sep 2026). **The product and its evaluation focus on the IT and computer-science (IT/CS) subset**: 15 journals selected from Lists 1 and 2 by subject area (section 2.6). Fields in the file: title (kk, ru or en, often mixed in one cell), publisher, scientific direction, inclusion date, order number and date. ISSN, language and website are **not in the file** and come from enrichment (OpenAlex, DOAJ, journal sites) or manual entry.
- Search: exact, partial, fuzzy, semantic; filters; status check.
- Import and update of registry data with change tracking.
- Deployment with Docker.

Out of scope

- Making or changing official accreditation decisions.
- Full article-level indexing, citation analysis.
- Automatic indexing of the thousands of international journals that enter List 1 automatically through Web of Science and Scopus. They are covered only by a rule note unless a legally usable list is obtained.
- Authentication and user accounts beyond a simple admin import key.

Engineering complexity

- Data engineering: irregular and changing source layout, normalisation of multilingual titles, ISSN validation, versioned imports with diff.
- Information retrieval: three retrieval methods and a fusion layer, tuned and evaluated on a purpose-built query set; multilingual (kk/ru/en) with limited tooling for Kazakh.
- Backend: PostgreSQL with FTS, `pg_trgm` and `pgvector` indexes, embedding service, API design, latency optimisation, containerised deployment.
- Frontend: responsive search interface with filters, tables, error and status handling, integrated with the API.
- Evaluation methodology: query set generation, relevance judgements, statistical comparison.

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2. Evidence of Relevance

2.1 Documentary evidence about the list (secondary sources, to be verified against the official order)

Finding	Source
Requirements for recommended journals were set out in a new edition by order No. 289 of the Minister of Science and Higher Education of 3 June 2025 (amending order No. 20 of the Minister of Education and Science of 12 January 2016), registered by the Ministry of Justice on 5 June 2025 under No. 36221; it enters into force ten calendar days after its first official publication	primary: text of the order in the Adilet legal-information system (read through the zakon.uchet.kz copy of the database, 24 Oct 2026); original record adilet.zan.kz/rus/docs/V2500036221; secondary: zako
The list has six components (point 3 of the requirements): scientific journals of level 1, 2 and 3; journals of national security and law-enforcement bodies; collections of international scientific conferences; list of trusted Kazakhstani publishers	same order text
List 1 includes all Web of Science Core Collection and Scopus journals automatically, plus listed trusted publishers; domestic List 1 journals need DOI for each article, IMRAD structure, blind review, licensed plagiarism check	same
A commercial guide gives counts "as of July 2026": List 1 46 domestic journals (plus thousands of international titles), List 2 164, List 3 19, and states the list is updated quarterly	a-articles.kz

Finding	Source
A 2024 Committee presentation described the earlier list as 152 journals in 18 directions; 174 before 2021, 147 in 2021	Committee presentation, via search summary
Experts estimated that at most 20% of previously included titles meet the new criteria	zakon.kz / search summary

Counts differ between sources and over time: the number of domestic journals changed between 2021 and 2026 and depends on the list level. A commercial guide gives 46 + 164 + 19 = 229; the official file of 15 Sep 2026 has 47 + 163 + 22 = 232 rows in Lists 1-3 (243 with the 11 military and security journals), see section 2.6.

What this means for the project:

- The data set is **small** (hundreds of rows, not millions), so the difficulty is not scale but quality of matching: multilingual, noisy queries, changing source, and trustworthy status information.
- The list **changes** (quarterly updates, criteria change), which justifies versioned imports and a visible "data last updated" date.
- The format of the list is **verified for one edition**: the file of 15 Sep 2026 is an xlsx with an irregular layout (section 2.6). That the Committee publishes the list in xlsx regularly is not yet confirmed (task DIP-9). This remains the main data risk (section 10).

2.2 Evidence about existing tools

Tool	What it offers	Source
Scopus Sources	free browse and search of sources by subject area, title, publisher or ISSN	Elsevier support
Web of Science Master Journal List	lookup by title, keyword or ISSN/eISSN; profile shows Core Collection indexes	CASRAI guide summary
Committee list	published as official documents/files; no search service found in our review	section 4
None of these tools answers "is this journal on the Kazakhstani Committee's list, at which level?" in one query across kk/ru/en. This is the practical gap of the project (details in section 4).		

2.3 Evidence from literature (verified, see section 3)

- Cyrillic and regional material breaks tools built for English: vanilla GROBID reaches F1 0.06 on Cyrillic headers (L1), and parsers need target-language data (L2, L3). The import of the official list is therefore tested on Kazakh and Russian rows, not assumed to work.
- No single database covers regional journals (L14, L15) and post-Soviet countries show low overall indexing in Scopus and Web of Science and their journals are mostly not international (L18); neighbouring countries built national indexes and selected lists (L17). A purpose-built registry of the Committee list fills a gap that the international sources leave.

2.4 User survey and follow-up discussions

The user needs are explored with an anonymous online survey (Google Forms) of potential users, followed by 3-5 short discussions with volunteers and the supervisor. Survey form: https://docs.google.com/forms/d/e/1FAIpQLScFAFdqwJFGHOFftV6nl4vmFLz6PFovjTx-XXTA9Z2oMPNAMA/viewform (English; kk and ru copies to follow). Until the responses are in, every user need in this report is an **assumption** (section 2.5).

Design

Item	Plan
Target groups	researchers and teachers; PhD and master's students; librarians and dissertation-council staff; journal editors
Sample	at least 30 responses, with at least 5 from each of the two largest groups
Distribution	university and faculty chats, department mailing lists, supervisor's contacts
Languages	form in kk, ru and en (English version is live; kk and ru copies to follow)
Period	24 Oct to 6 Nov 2026 (reminder after one week)
Privacy	anonymous: no names or e-mails in the form; a separate optional field for follow-up contact, stored outside the repository; no personal data committed
Use of results	counts and percentages per closed question; open answers coded into themes; every finding mapped to a requirement id (section 6); real search queries added to the evaluation set (section 9.2)

Questionnaire (to paste into the form)

1. Your role (researcher/teacher, PhD student, master's student, librarian or council staff, editor, other).
2. Your field (IT and computer science, other technical, natural sciences, humanities and social sciences, medicine, other).
3. How often do you check whether a journal is on the Committee's list? (never, yearly, monthly, weekly or more)
4. Where do you check it today? (official file, university site, Scopus or Web of Science, search engine, colleagues, other)
5. How long does a typical check take? (under 1 minute, 1-5 minutes, more than 5 minutes)
6. Have you ever got a wrong or unclear answer about a journal's status? (yes/no; if yes, what happened)
7. What do you usually search by? (title, abbreviation, ISSN, publisher, topic; several allowed)
8. In which languages do you type journal names or topics? (kk, ru, en; several allowed)
9. Do you often make typos or use abbreviations or transliteration when searching? (never, sometimes, often)
10. Do you search for journals by topic, such as "journals on information security"? (never, sometimes, often)
11. What must the answer show? Rate 1-5 in a grid: status, list level (1/2/3), subject area, date of inclusion or removal, link to the official order, ISSN, website.
12. Would you trust the result without opening the official file? (yes, only if the source and date are shown, no)
13. Write one or two real queries you would type (title, ISSN or topic).
14. What is the most annoying part of checking journals now? (open text)
15. Optional: may we contact you for a 15-minute follow-up? (leave an address; it is not stored in the repository)

Follow-up discussions (3-5, 20 minutes each)

#	Role	Why	Contact / date	Status
U1	PhD student or young researcher	how status is checked today; typical queries	in progress	in progress
U2	Teacher or researcher who publishes regularly	decision before submission; which attributes matter	in progress	in progress
U3	Librarian or scientometrics specialist	verification routines, error cases, official sources	in progress	in progress
U4	Dissertation council secretary or PhD office	which level is required for defence; proof needed	in progress	in progress
U5	Supervisor (client proxy)	scope, data source, priorities	in progress	in progress

Note template per discussion:

```
Interviewee role / org:      Date / format:
Key facts stated (verbatim where possible):
Example queries given (language):
Requirement implied (-> FR/NFR id in section 6):
Changes our assumptions? (which):
Follow-ups:
```

Results (the form is open and the discussions are in progress; results are filled after the form closes on 6 Nov 2026, no figures are reported before that)

Question	Result	Requirement affected
number of responses, group mix	in progress	
Q3-Q6: frequency, tools, time, errors	in progress	FR-16, NFR-01
Q7-Q9: search by, languages, typos	in progress	FR-05 to FR-09
Q10: topic search	in progress	FR-07
Q11: fields the answer must show	in progress	FR-04, FR-14
Q12: trust and source display	in progress	FR-16

2.5 Verified findings versus assumptions

Statement	Status	Basis	How to verify
The 2025 order created three list levels	**Verified** in the order text (No. 289 of 3 June 2025, registered 5 June 2025 under No. 36221, point 3 of the requirements lists levels 1, 2 and 3)	Adilet database text, read 24 Oct 2026 through a copy of the database	cite the order by number; re-check when the order is amended (the list of amendments is shown on the Adilet page)
About 229 domestic journals in lists 1-3	**Checked** against the file: 232 rows in lists 1-3 (47 / 163 / 22) plus 11 in list 4; commercial counts are close but outdated	official file of 15 Sep 2026	re-count when a new edition appears
The list is updated quarterly	**Unverified**	a-articles.kz	check official file history
Scopus and WoS offer title/ISSN search but not the Committee status	Verified for the tools' own search features	vendor documentation	try both with sample journals
Researchers find the list inconvenient to search	Assumption (supervisor's justification)	none yet	survey Q3-Q6 and discussions U1-U4 (section 2.4)
Users query in kk, ru and en with typos and abbreviations	Assumption	none yet	survey Q7-Q9, Q13
Hybrid retrieval beats single methods on this data	**Hypothesis**	no direct evidence for short multilingual journal titles; to be measured	evaluation in section 9
Data file is machine-readable	**Verified** for the edition of 15 Sep 2026 (xlsx, one sheet per list, merged header text, no ISSN column)	the list file	confirm that the Committee publishes xlsx regularly (DIP-9)

2.6 Analysis of available data: the official list file

Source: the Committee's list file "КОКСНВО от 15 сентября 2026 г." (xlsx, sheets for Lists 1, 2, 3, list 4 for military and security university journals, and the team's IT/CS selection).

Sheet	Rows with a journal	Rows whose direction mentions IT, computer science, informatics or mathematics (rough keyword match)
List 1	47	13
List 2	163	14
List 3	22	1
List 4 (military, security)	11	0
Total	**243**	

Columns: order number, title, publisher, scientific direction, date or issue from which the journal is in the list, order date and number. There is no ISSN, language, website or status column.

Irregularities found for the import module (FR-01, FR-02):

- Sheet text, headers and subjects are mostly in Kazakh; titles mix Kazakh, Russian and English in one cell, sometimes with a former title ("бұрынғы атауы").
- Numbers are stored as text or floats ("13.0", "45."); rows are not sorted; blank rows and a row split across two lines occur (journal 110).
- Inclusion cells use free text ("наурыз 2021 жыл", "2026 жылғы қаңтар айынан бастап"), and one cell can hold several orders (a journal added in 2024 and amended in 2025).
- Subject is free text with many comma-separated directions, not a controlled vocabulary.
- One IT/CS row (ҚазТБУ Хабаршысы) has no number.

IT/CS subset (15 journals). Selected by direction containing computer science, information and communication technologies, informatics, or automation and IT:

# in file	Journal	Publisher	Level
13	Eurasian Journal of Mathematical and Computer Applications	L.N. Gumilyov ENU	1
45	Herald of the Kazakh-British Technical University	KBTU	1
23	Bulletin of the L.N. Gumilyov ENU, Mathematics, Computer Science	L.N. Gumilyov ENU	2
38	Bulletin of the National Engineering Academy of the RK	National Engineering Academy	2
42	Academic Scientific Journal of Computer Science Kazakhstan	Central Asia Academic Scientific Center LLP	2

# in file	Journal	Publisher	Level
54	Proceedings of the University (Karaganda Technical University)	KTU	2
135	Journal of Problems in Computer Science and Information Technologies	Al-Farabi KazNU	2
136	Bulletin of the Civil Aviation Academy	Civil Aviation Academy	2
48	Bulletin of Abai KazNPU, Physics-Mathematics	Abai KazNPU	2
85	Scientific Journal of Astana IT University	Astana IT University	2
-	Bulletin of Kazakh University of Technology and Business (ҚазТБУ)	KazUTB	2
110	Bulletin of Shakarim University, Technical Sciences	Shakarim University	2
111	Bulletin of the M. Tynyshpayev Kazakh Academy of Transport and Communications	KazATC	2
139	International Journal of Information and Communication Technologies	International IT University	2
155	EKTU Journal of Information and Communication Sciences	D. Serikbayev EKTU	2

Levels were read by matching titles to the list sheets (journals 13 and 45 are in List 1, the others in List 2); re-check them in the import test (DIP-9). Titles are translated for readability; the file keeps the original.

Consequences for the design:

1. ISSN lookup (FR-09) needs a second source. The core fields come from the file; ISSN and eISSN are added by an enrichment step (OpenAlex or DOAJ API, journal sites) and by manual entry, with a flag showing where each value came from.
2. The evaluation set (section 9) has two parts: the full registry (243 rows) for title, typo and cross-language queries, and the IT/CS subset for topic queries ("journals on machine learning", "cybersecurity"), where semantic search should help most.
3. Subject strings must be normalised into a small controlled list before topic filters work.

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3. Literature Review

Requirement: at least 12 meaningfully used sources, at least 8 scientific, mostly from the last five years, with foundational works kept. This review has 26 sources: 11 open-access papers selected by a structured database search (L1-L11, search log in Appendix A), 7 added in October 2026 by targeted web search on indexes and coverage (L12-L18) and 8 added on 24 October 2026 on retrieval, fusion, multilingual embeddings and Kazakh language resources (L19-L26). Bibliographic data of L12-L26 were checked against DOI, publisher, ACL Anthology or arXiv pages. 25 are scientific and 1 is a tool paper (L12). 17 are from 2021 or later. Full BibTeX: lit/references.bib.

The review asks what published work says about five things the system depends on: (1) reading metadata and lists in Cyrillic and other non-English scholarly material, (2) how scholarly indexes and journal lists are built and how far they cover regional journals, (3) which text models suit scientific and multilingual text, (4) how rankings from several retrievers are fused, and (5) what is known about retrieval and language resources for Kazakh. Themes 1 and 2 mostly concern the import step and the case for the registry; themes 3 to 5 concern the search core and the research question.

3.1 Source table

#	Source	Year	Type	Focus	Used for
L1	Krause, Shapiro, Saier, Färber, SDP workshop (ACL)	2021	scientific	metadata extraction on Cyrillic PDFs	evidence that Western tools fail on Cyrillic; import risk
L2	Choi, Yoon, Hyun, Lee, PLOS ONE	2023	scientific	multilingual reference corpus, BERT-REF	multilingual parsing, annotation quality
L3	Kong, Yoon, Seol, Lee, Choi, IEEE Access	2023	scientific	Korean journals, text plus 2D boxes	national-corpus approach, layout features
L4	Lo, Wang, Neumann, Kinney, Weld, ACL (S2ORC)	2020	scientific, foundational	open full-text corpus pipeline	modular ingestion pattern
L5	Nicholson et al., Quantitative Science Studies (scite)	2021	scientific	citation index at production scale	how an index is built from extracted data
L6	Choi, Kong, Yoon, Computers, Materials & Continua (LAME)	2022	scientific	layout-aware metadata extraction	layout features across journal templates
L7	Alyafeai, Al-Shaibani, Ghanem, Findings of EMNLP (MOLE)	2025	scientific	LLM metadata extraction	LLM option for irregular tables
L8	Blecher, Cucurull, Scialom, Stojnic, ICLR 2024 / arXiv (Nougat)	2023	scientific	vision-based document parsing	heavy alternative, rejected
L9	Cohan, Feldman, Beltagy, Downey, Weld, ACL (SPECTER)	2020	scientific, foundational	citation-informed document embeddings	embeddings for journal and topic text
L10	Saier, Krause, Färber, JCDL (unarXive 2022)	2023	scientific	full text linked to OpenAlex	linking records to an open graph
L11	Beltagy, Lo, Cohan, EMNLP (SciBERT)	2019	scientific, foundational	scientific vocabulary	domain language models
L12	Lopez, ECDL (GROBID)	2009	tool paper	CRF header and reference extraction	baseline parser
L13	Tkaczyk et al., IJDAR (CERMINE)	2015	scientific	born-digital PDF metadata, average F-score 77.5%	second baseline parser
L14	Visser, van Eck, Waltman, Quantitative Science Studies	2021	scientific	Scopus, Web of Science, Dimensions, Crossref, Microsoft Academic compared	coverage limits of databases
L15	Martín-Martín, Thelwall, Orduna-Malea, Delgado López-Cózar, Scientometrics	2021	scientific	citation coverage of six databases	why one database cannot answer a national-list question
L16	Priem, Piwowar, Orr, arXiv (OpenAlex)	2022	scientific (preprint)	open index of works, venues, institutions	optional enrichment of journal records
L17	Moskaleva, Pislyakov, Sterligov, Akoev, Shabanova, Scientometrics (Russian Index of Science Citation)	2018	scientific	national citation index and journal selection in a neighbouring country	analogue of a national list
L18	Alimova, Brumshteyn, European Science Editing	2020	scientific	coverage of papers of Russia and post-Soviet countries by Scopus and Web of Science, languages, productivity	regional coverage gap
L19	Cormack, Clarke, Büttcher, SIGIR	2009	scientific, foundational	Reciprocal Rank Fusion (RRF), k = 60	fusion method of the search core (FR-08)
L20	Bruch, Gai, Ingber, arXiv (published in ACM TOIS)	2022	scientific (preprint)	fusion functions for hybrid lexical and semantic retrieval: RRF versus convex combination	ablation H2, tuning of weights
L21	Chen, Xiao, Zhang, Luo, Lian, Liu, Findings of ACL (BGE M3-Embedding)	2024	scientific	multilingual embeddings, 100+ languages, dense, sparse and multi-vector	embedding model A (section 5.3)
L22	Wang, Yang, Huang, Yang, Majumder, Wei, arXiv (multilingual E5)	2024	scientific (technical report)	multilingual E5 small, base and large	embedding model B (section 5.3)
L23	Reimers, Gurevych, EMNLP	2020	scientific, foundational	multilingual sentence embeddings by knowledge distillation	cross-language matching, MiniLM family option
L24	Muennighoff, Tazi, Magne, Reimers, EACL (MTEB)	2023	scientific	benchmark of embedding models, 58 datasets, 8 tasks, 112 languages	how to compare embedding models; limits of public benchmarks
L25	Yeshpanov, Efimov, Boytsov, Shalkarbayuli, Braslavski, LREC-COLING (KazQAD)	2024	scientific	Kazakh open-domain QA dataset with passage-level relevance judgements	the only retrieval baseline for Kazakh found
L26	Yeshpanov, Khassanov, Varol, LREC (KazNERD)	2022	scientific	Kazakh named entity recognition dataset	state of Kazakh language resources

3.2 Structured analysis of the core publications (L1-L11)

Metrics are as reported by the authors; we did not reproduce them.

Source	Research problem	Research method	Dataset / sample	Technology or approach	Evaluation metrics	Main findings	Reported limitations
L1 Krause, Shapiro, Saier, & Färber (2021)	Off-the-shelf scholarly metadata extractors (vanilla GROBID) are trained on English/Latin typography and experience severe performance collapse when processing Cyrillic academic PDFs.	Weakly supervised bootstrapping pipeline combining heuristic rules, character n-gram language detection, pdftotext spatial stream extraction, and deep sequence labeling.	15,553 Cyrillic open-access academic publications from CORE collection across Ukrainian (11,708), Russian (3,786), and Bulgarian (54).	pdftotext spatial stream extraction, BiLSTM sequence tagging with fastText embeddings, and retrained GROBID (CRF).	Precision, Recall, Macro and Micro F1-score across header metadata fields (Title, Author, Misc).	Vanilla GROBID achieved an F1 of only 0.06 on Cyrillic headers; retrained GROBID improved F1 to 0.83; standalone BiLSTM achieved F1 = 0.90 (precision 0.88, recall 0.91).	Does not explicitly address 2D visual layout coordinates; requires clean digital text layers and fails on scanned bitmap PDFs.

Source	Research problem	Research method	Dataset / sample	Technology or approach	Evaluation metrics	Main findings	Reported limitations
L2 Choi, Yoon, Hyun, & Lee (2023)	Standard reference parsers (GROBID, CERMINE) are limited to English references and fail to extract diverse non-Western metadata fields from multilingual reference lists.	Corpus construction with rigorous annotation guidelines followed by transfer learning using BERT-based sequence labeling (BERT-REF).	3,815,987 multilingual references (mainly English and Korean) labeled across 13 distinct bibliographic metadata fields.	BERT-based transfer learning token classification (BERT-REF) trained on BIO-tagged sequences, benchmarked against GROBID.	Token-level and entity-level Precision, Recall, and F1-score; inter-annotator agreement kappa.	BERT-REF achieved 99.83% token F1 score across 13 reference fields; high inter-annotator agreement (kappa = 0.941), significantly outperforming vanilla GROBID on multilingual references.	Performance drops on references with non-standard abbreviations, irregular punctuation delimiters, or unindexed regional conference proceedings.
L3 Kong, Yoon, Seol, Lee, & Choi (2023)	Scarcity of open annotated academic corpora for national research ecosystems and inability of conventional text-only parsers to utilize 2D spatial layouts of academic article headers.	Large-scale automated corpus construction followed by fine-tuning domain-adapted BERT integrated with 2D spatial bounding-box coordinates.	315,320 PDF papers belonging to 503 Korean academic journals across 12 metadata types.	SciBERT-ME-J model combining text tokens with 2D bounding-box spatial coordinates (x0, y0, x1, y1) extracted via PDFMiner.	Field-level F1-score, Precision, and Recall on Title, Author, Affiliation, Abstract, and Keywords; inference latency.	SciBERT-ME-J achieved an outstanding F1 score of 99.36% on header metadata extraction, demonstrating that 2D layout coordinates prevent column reading-order errors.	Manual inspection was required for test sets; extraction accuracy drops on papers deviating from standard journal layout templates.
L4 Lo, Wang, Neumann, Kinney, & Weld (2020)	Fragmented, proprietary, and paywalled scholarly corpora lacking structured full-text, resolved citation links, and open machine-readable formats for downstream NLP tasks.	Large-scale distributed automated ingestion, normalization, PDF-to-TEI XML extraction using customized parser pipeline, and citation graph resolution.	S2ORC corpus comprising 81.1 million English academic paper metadata records and 8.1 million full-text open-access papers.	Distributed pipeline integrating SCIENCEPARSE v3.0.0 and GROBID v0.5.5, parsing PDF into TEI XML/JSON with resolved inline citation links.	In-text citation linking precision and coverage, TEI XML parsing validity, multi-disciplinary corpus coverage.	Constructed the largest open research corpus with parsed inline citations, establishing modular pipeline architecture for scalable citation indexing.	Ingestion pipeline was designed strictly for English publications; extraction fails or exhibits high corruption rates on non-Latin fonts and non-standard national journal formats.
L5 Nicholson et al. (2021)	Traditional academic citation indices treat citations as flat, unweighted numerical counts, failing to provide context or distinguish between supporting, mentioning, or contrasting citations.	Large-scale deep learning classification pipeline built atop publisher PDF streams to extract in-text citation contexts and classify author intent.	Over 23 million full-text scientific articles and a production database of more than 800 million classified citation statements.	Extended GROBID engine for extracting citation sentences, coupled with deep learning neural classifier for tri-class intent classification (supporting, disputing, mentioning).	Classification precision, recall, and F1-score across citation context categories; citation parsing accuracy.	Demonstrated that context-aware smart citation indexing is operationally viable at massive scale (800M+ statements), providing qualitative depth to research evaluation.	Substantial compute overhead required for full-text processing; reliance on publisher PDF layout consistency; reduced accuracy when citation statements rely on complex multi-sentence argumentation.
L6 Choi, Kong, & Yoon (2022)	Diverse formatting and irregular multi-column layouts across journals disrupt standard left-to-right text parsers, conflating header metadata fields.	Development of LAME (LAyout-aware Metadata Extraction) framework integrating PDFMiner automatic layout analysis with pre-trained transformer modeling.	Large corpus of diverse academic journal articles with varying layout formats from domestic Korean publishers.	LAME framework equipped with PDFMiner 2D coordinate extraction and Layout-MetaBERT for layout-aware sequence modeling.	Macro-F1, Precision, and Recall across 5 core metadata fields (title, abstract, author name, affiliation, keywords).	Layout-MetaBERT achieved Macro-F1 = 93.27% across diverse layout formats, confirming that 2D layout features prevent field boundary confusion in multi-column formats.	High memory and GPU processing requirements; sensitive to irregular multi-column splits and floating author footers.
L7 Alyafeai, Al-Shaibani, & Ghanem (2025)	Manual annotation of scientific metadata cannot keep pace with the exponential growth of scholarly literature, while traditional rule-based tools lack generalizability.	MOLE framework leveraging Large Language Models (LLMs) with schema-driven prompting, multi-format processing, and automated validation loops.	Benchmark of 126 scientific papers across 3 formats (PDF, Markdown, text) evaluated on 32 distinct metadata attributes across 7 LLMs.	Prompt-engineered LLMs (open-weight and proprietary) with structured JSON decoding and self-consistency schema validation guards.	Schema compliance, exact match, attribute-level F1-score, and hallucination frequency.	Modern LLMs exhibit strong zero-shot extraction capability across multilingual documents (F1 > 0.90), but require schema validation to ensure consistency.	High inference latency and computational expense make LLMs impractical for million-paper batch ingestion without hybrid filtering; risk of hallucinated metadata if prompt constraints fail.
L8 Blecher, Cucurull, Scialom, & Stojnic (2023)	Programmatic PDF text layers frequently lose semantic information, scramble reading order, and corrupt mathematical expressions in scientific documents.	Visual document understanding paradigm treating academic document parsing as an end-to-end vision-to-sequence translation task without relying on underlying text layers.	1,748,201 articles released on arXiv paired with LaTeX source files, plus PubMed Central open-access articles.	Nougat architecture: Swin Transformer base encoder (96 DPI raster input) + mBART decoder (10 layers, 350M parameters) generating Markdown/LaTeX.	Normalized Edit Distance, BLEU, and formula reconstruction accuracy.	Demonstrates high-fidelity parsing of complex scientific text and math formulas directly from page raster images, bypassing damaged font streams completely.	Autoregressive generation is computationally slow (several seconds per page); model can occasionally get stuck in repetitive token generation loops; requires GPU acceleration.
L9 Cohan, Feldman, Beltagy, Downey, & Weld (2020)	Standard BERT models produce suboptimal document-level embeddings because they only encode local text tokens without leveraging inter-document citation graph topology.	Citation-informed pre-training objective where paper embeddings are trained to pull cited papers closer and push non-cited papers further away in vector space.	146K query papers (26.7M tokens) and 684K training triples with hard and easy negatives from Semantic Scholar.	SPECTER transformer encoder trained using triplet margin loss on citation links; requires no citation graph structure during inference.	Accuracy, MAP, and F1-score across the SCIDOCS evaluation benchmark (7 document-level tasks: classification, citation prediction, recommendation).	Consistently outperforms SciBERT and BERT baselines on all 7 SCIDOCS document-level tasks without task-specific fine-tuning; eliminates the need for graph inputs at inference time.	Effectiveness depends on dense citation networks, which may be sparse in emerging or localized regional research domains.
L10 Saier, Krause, & Färber (2023)	Existing scholarly full-text corpora suffer from limited domain and time coverage, incomplete citation networks, and lossy representation of mathematical notations.	Construction of unarXive 2022 via an automated pipeline pre-processing arXiv papers into structured full-text and linking citations to open knowledge graphs.	1.9 million publications from arXiv across multiple scientific disciplines spanning 32 years (1991–2022).	Automated pipeline integrating LaTeXML and GROBID with persistent OpenAlex entity resolution.	In-text citation marker resolution rate, citation graph completeness, database query latency.	Successfully linked 64,547,944 in-text citation markers directly to OpenAlex, creating a complete open citation graph with structured full text and LaTeX math formulas.	Pipeline relies heavily on the presence of LaTeX source files in arXiv; pure PDF fallback parsing exhibits higher reference extraction error rates.
L11 Beltagy, Lo, & Cohan (2019)	General-domain BERT models exhibit only 42% token overlap with scientific terminology, causing substantial out-of-vocabulary errors on scholarly document processing tasks.	Pre-training a domain-specific BERT architecture from scratch using a specialized scientific vocabulary on a large corpus of scientific literature.	1.14 million scientific papers (3.17 billion tokens; 18% computer science, 82% biomedical) from Semantic Scholar.	SciBERT: BERT-base architecture trained with custom 30K WordPiece vocabulary (SciVocab) built using SentencePiece.	F1-score across 8 benchmark NLP tasks: named entity recognition (NER), relation extraction, dependency parsing, text classification.	Consistently outperforms BERT-base across all 8 scientific NLP benchmarks (+2.5 to +4.0 F1); established that domain-specific vocabulary is foundational for scientific text processing.	Pretrained exclusively on English-language publications; lacks multilingual/Cyrillic tokens, requiring fine-tuning or adaptation for national non-Latin corpora.

3.3 Added sources (L12-L26)

Source	What it contributes	Limitation for this project
L12 Lopez 2009	GROBID, the usual open baseline for header and reference extraction (CRF)	tool paper; trained on Latin-script data (see L1)
L13 Tkaczyk et al. 2015	CERMINE, open-source extraction from born-digital PDFs, modular, average F-score 77.5%	English-oriented; older than the transformer methods in L2, L3, L6
L14 Visser et al. 2021	compares Scopus, Web of Science, Dimensions, Crossref and Microsoft Academic on document coverage over time, document types and disciplines, and on citation links; recommends broad coverage with flexible filters	studies 2008-2017 and article records, not journal lists
L15 Martín-Martín et al. 2021	3,073,351 citations to 2,515 highly cited documents across six sources: Google Scholar found about 88%, Microsoft Academic 60%, Dimensions 54%, COCI 28%	English-language, highly cited documents only
L16 Priem et al. 2022	OpenAlex: open index (CC0) of 209M works, authors, venues, institutions; REST API and data dump	preprint; accuracy of venue and ISSN data for Kazakhstani journals not checked
L17 Moskaleva et al. 2018	describes the Russian Index of Science Citation (launched 2005): about 25 million documents in the database, a Web of Science-hosted selection of 652 journals, citation contexts from full text of 4,391 journals	describes Russia, not Kazakhstan; figures are those of the paper and may be out of date
L18 Alimova & Brumshteyn 2020	compares post-Soviet countries by output, share of papers in Scopus and Web of Science, and language; reports low overall values (about 1.04 Scopus and 0.87 Web of Science papers per researcher on average) and that journals mostly publish in national languages (often Russian) or English, with few foreign board members, so most cannot be called international	abstract only was read; full-text figures not checked; covers papers and countries, not national lists
L19 Cormack et al. 2009	RRF: sum of 1 / (k + rank) over rankings, k = 60; reported to beat the best single system and Condorcet fusion in TREC experiments and on LETOR 3	2009 web-search and learning-to-rank data, long documents; not short multilingual titles
L20 Bruch et al. 2022	compares fusion functions for lexical plus semantic retrieval; reports that a convex combination of normalised scores is at least as effective as RRF and needs only a small tuning set, while RRF is sensitive to its parameters (as reported by the authors; we have not reproduced it)	English benchmarks; we read the preprint, not the journal version

Source	What it contributes	Limitation for this project
L21 Chen et al. 2024	M3-Embedding: one model for 100+ languages, three retrieval modes (dense, sparse, multi-vector), inputs up to 8,192 tokens	large model, CPU latency unknown for our deployment; Kazakh quality not reported for registry-style titles
L22 Wang et al. 2024	multilingual E5: small, base and large models trained by contrastive pre-training on about 1 billion multilingual pairs; the small and base sizes trade quality for efficiency	same: nothing on very short proper-noun titles
L23 Reimers & Gurevych 2020	trains a multilingual student to reproduce an English sentence-embedding model, so translations land close together in vector space	needs parallel data per language; quality for Kazakh not shown here
L24 Muennighoff et al. 2023	MTEB: 58 datasets, 8 task types, 112 languages; shows no embedding model wins on all tasks	public benchmark; cannot replace testing on our own queries
L25 Yeshpanov et al. 2024	KazQAD: just under 6,000 questions and nearly 12,000 passage-level relevance judgements over more than 800,000 Kazakh Wikipedia passages; baseline retrievers reach NDCG@10 0.389 and MRR 0.382	encyclopaedic passages, not journal titles
L26 Yeshpanov et al. 2022	KazNERD: 112,702 sentences and 136,333 annotations in 25 entity classes, best model 97.22% exact-match F1	news text; shows resources exist for tagging, not for retrieval of titles

3.4 Critical synthesis

What is known.

1. Extraction tools trained on English fail on Cyrillic unless retrained: vanilla GROBID reaches F1 0.06 on Cyrillic headers and 0.83 to 0.90 after retraining (L1). Large annotated corpora in the target language lift reference parsing to above 99% token F1 (L2, L3). Layout features reduce field confusion on multi-column pages (L3, L6, agreeing).
2. LLM and vision-based extractors work without training but cost latency and compute, and need validation against hallucinated fields (L7, L8).
3. Production indexes are modular pipelines with entity linking to open metadata (L4, L5, L10), and open metadata sources such as OpenAlex can supply venue records (L10, L16).
4. No single database covers all journals: coverage differs strongly between Scopus, Web of Science, Dimensions, Google Scholar and open sources (L14, L15), and post-Soviet countries show low overall indexing in the two international indexes, with journals mostly publishing in national languages (L18). Neighbouring countries solved this with national indexes and selected journal lists (L17).
5. Domain-specific text models beat general ones on scientific language (L11, L9), but the published ones are English-centred.
6. RRF is a simple, parameter-light way to fuse rankings (L19). Later work reports that a tuned convex combination can match or beat it and that RRF depends on its parameters (L20). Both are therefore kept in the design: RRF as the proposed fusion, convex combination as an ablation (H1 versus H2).
7. Multilingual embedding models exist at several sizes and cover 100+ languages (L21, L22, L23), and benchmarks show that the best model depends on the task (L24). The choice for this project is therefore made by measurement (section 5.3), not by reading leaderboards.
8. Kazakh is a low-resource language: retrieval resources are few (the Kazakh QA dataset L25 gives retrieval baselines of about 0.38 to 0.39 on encyclopaedic passages) and annotated resources are mostly for tagging (L26).

Where the evidence is weak for this project.

- The best extraction numbers come from Korean or English corpora (L2, L3, L6) and are upper bounds for Kazakhstani material; Kazakh is absent from every extraction study. For retrieval, Kazakh appears only in L25 and L26, and none of the fusion and embedding papers (L19-L24) reports results on short journal titles in Kazakh.
- Most of the early sources (L1-L13) target article-level extraction. The input here is a spreadsheet of about 240 journals (section 2.6), so the extraction literature mainly informs risk R1 and the design of the import step; the core search rests on L19-L26.
- The fusion and embedding papers use long documents or passages. Journal titles are short proper nouns, so published gains for RRF or for any embedding model are not expected to transfer unchanged; this is the reason for the pilot in section 9.7.
- The coverage studies (L14, L15) compare databases of articles, not lists of journals approved by a regulator.

Gap. No work was found on a searchable registry of a national list of recommended journals, nor on status lookup against such a list for Kazakhstan, nor on hybrid retrieval over short multilingual journal titles that includes Kazakh. The nearest analogue is the Russian national index (L17), described by its operators. Absence in our queries is not proof of absence (Appendix A).

3.5 What the project takes from the literature

Decision	Taken from
Treat the official list as semi-structured Cyrillic data: parse it with tested rules, keep a manual CSV fallback, and test on Cyrillic and Kazakh (risk R1)	L1, L2, L12, L13
Use rule-based parsing first; consider an LLM with schema validation only for irregular rows, never as an unchecked step	L7
Do not use a vision model (GPU, seconds per page) for a 240-row table	L8
Store the registry as a versioned, linked metadata index with an enrichment step for ISSN and venue data (OpenAlex as optional source)	L4, L5, L10, L16
Justify a purpose-built registry because no single database answers the Committee-list question	L14, L15, L18, L17
Choose text models for scientific and multilingual text by testing on our own data, since published domain models are English-centred	L9, L11
Fuse the three rankings with RRF, k = 60 as default and tuned on the development split; keep a convex combination as ablation	L19, L20
Shortlist BGE M3 and multilingual E5 (small or base) as embedding candidates; keep the MiniLM family as a lighter option; decide by measurement because benchmark winners differ by task	L21, L22, L23, L24
Add a trigram retriever and a transliteration table as safety net for Kazakh, and report every metric per language	L25, L26
Report results per language and record the Kazakh gap	L1, L2, L3

3.6 Count

Scientific sources: 25 (L1-L11, L13-L26). Tool paper: 1 (L12). Total 26. From 2021 or later: 17 (L1, L2, L3, L5, L6, L7, L8, L10, L14, L15, L16, L20, L21, L22, L24, L25, L26). Older foundational works kept on purpose: L4, L9, L11, L13, L19, L23.

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4. Existing System Comparison

Five systems on nine criteria. Cells say **Yes / Partial / No / Not verified** with the evidence. "Not verified" means we found no source; it is not a negative claim. Evidence was gathered on 21 Oct 2026 from vendor documentation and search summaries; no tool was tried hands-on yet (task DIP-8, section 4.5).

4.1 Systems

ID	System	Type	Evidence
S1	Committee list as published (legal-information systems and university copies)	official document, three levels	Adilet, prg.kz copy as of July 2024, zakon.kz
S2	Scopus Sources	commercial source list, free browse	Elsevier support
S3	Web of Science Master Journal List	commercial journal list, free lookup	CASRAI guide summary
S4	DOAJ (Directory of Open Access Journals)	open directory with API	third-party API guides and a DOAJ test-site copy of its docs; v4 syntax not confirmed
S5	OpenAlex Sources	open scholarly index, journals as "sources"	OpenAlex docs

4.2 Criteria

ID	Criterion	Why it matters
C1	Knows the Committee list status and level	the central question of the project
C2	Search by title	basic need
C3	Search by ISSN	precise lookup, handles renamed journals
C4	Typo / fuzzy tolerance	users type imperfect names
C5	Kazakh, Russian and English titles	local journals use all three
C6	Semantic or topic search	"journals on information security"
C7	Filters (level, subject, language)	narrowing results
C8	Open API or downloadable data	integration, reproducibility
C9	Update visibility (date, history)	the list changes

4.3 Comparison

Criterion	S1 Committee list	S2 Scopus Sources	S3 WoS MJL	S4 DOAJ	S5 OpenAlex Sources
C1 Committee status	**Yes**, it is the source. Levels 1-3 per the 2025 order	**Partial.** Scopus-indexed journals enter List 1 automatically, but Scopus does not show the Committee status	**Partial.** Same: WoS Core Collection journals enter List 1 automatically; no Committee status shown	**No**	**No**
C2 Title search	**Partial.** Document or table, text search depends on the viewer; no search service found	**Yes.** Title search	**Yes.** Title and keyword	**Yes** (free-text journal search per API guides)	**Yes.** `search` parameter
C3 ISSN search	**Partial.** Visible if ISSN is in the table	**Yes**	**Yes**, ISSN and eISSN	**Yes** (ISSN lookup per guides)	**Yes.** `issn_l` and `issn` filters (documented in reference and openalexR examples)
C4 Fuzzy / typo tolerance	**No** (document)	**Not verified**	**Not verified**	**Not verified** (Elasticsearch based, per docs syntax)	**Not verified**
C5 kk / ru / en	**Yes** for the titles as printed; no cross-language search	**Partial.** Titles as indexed, mostly Latin script	**Partial.** Same	**Partial.** Titles as supplied	**Partial.** Titles as supplied; Kazakh coverage not verified
C6 Semantic / topic search	**No**	**Partial.** Browse by subject area	**Partial.** Keyword lookup	**Partial.** Subject fields	**Partial.** Topic data exist; quality for this use not verified
C7 Filters	**Partial.** Level and subject are sections or columns, no interactive filter	**Yes.** Subject area and others	**Partial.** Not verified in detail	**Yes** (fields)	**Yes.** Many filters
C8 Open API / data	**Partial.** Legal text and files; machine-readable form not verified	**No** public API for this feature (not verified)	**No** (not verified)	**Yes**, public API, rate limits unclear	**Yes**, open API; CC0 licence reported in its API spec
C9 Update visibility	**Partial.** Edition date ("as of July 2024" on one copy); quarterly updates claimed by a commercial source	**Not verified**	**Not verified**	**Not verified**	**Not verified**

4.4 Conclusions

1. The only authoritative source of status (S1) is a document, with no fuzzy, cross-language or semantic search found, and an update trail only by edition date.
2. The international tools (S2-S5) search titles and ISSNs well, but none reports the Kazakhstani Committee status, which is the question users ask.
3. Public APIs (S4, S5) could enrich records (ISSN, publisher), but are not a substitute.
4. A system that imports S1, normalises it, and adds typo-tolerant, multilingual and semantic search with a status answer fills a concrete gap.

4.5 Limits and next steps

- Most cells rely on documentation, not on our own trials. By milestone M1, run ten sample journals through S2, S3, S5 and the official file and record what each returns (task DIP-8).
- The format in which the Committee publishes S1 is verified for one edition only (xlsx of 15 Sep 2026, section 2.6). Legal-information systems and university sites also carry the list as documents and PDF copies; whether xlsx is published regularly is checked in task DIP-9.
- Scopus and Web of Science were included only as search-feature comparisons; their data are licensed and are not a source for the system.

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5. Technology Comparison

Core decisions are scored against the requirement ids in section 6 (FR-xx functional, NFR-xx non-functional). Scores: ++ strong fit, + fit, 0 neutral, - weak, -- unfit. Facts come from the documentation and papers cited in section 3; **speed and relevance of each option have not been measured yet**, so choices are provisional and are confirmed by the benchmarks in section 9. Decision records are in Appendix B.

5.1 Decision 1: retrieval strategy

Criterion	A. FTS only	B. Vector only	C. Hybrid FTS + fuzzy + vector, fused by RRF
Exact and partial titles, ISSN (FR-05, FR-09)	++	- short proper nouns embed poorly	++
Typos and transliteration (FR-06)	-- stemming cannot fix typos	+ embeddings tolerate some noise	++ with trigram component
Topic and cross-language queries (FR-07)	--	++ multilingual models	++
Evidence for this project	0 to be measured (baseline M1)	0 to be measured (M4)	0 to be measured (H1, H2, section 9)
Complexity and latency (NFR-02)	++	+ embedding at query time	0 three queries plus fusion
Explainability for users	++	-	+ show which method matched
Decision: C, with A, trigram-only and B kept as baselines; fusion by RRF (k = 60, configurable; Cormack et al., L19) plus a convex-combination variant as ablation (Bruch et al., L20). The supervisor's document also specifies hybrid FTS plus vector search with RRF (ADR-001).			

5.2 Decision 2: search and storage engine

Criterion	A. PostgreSQL (FTS + `pg_trgm` + `pgvector`)	B. Elasticsearch / OpenSearch	C. SQLite FTS5 + external vector index (FAISS)
One system for data and search (FR-04, FR-05-08)	++ single database, one transaction for imports	0 second system to sync	- two stores
FTS, fuzzy, vector all available (FR-05-07)	++ built in or extensions (pg_trgm docs, pgvector docs)	++ native	+ FTS5 yes, fuzzy weak
Fit with registry size (hundreds of rows)	++ exact vector scan is enough, HNSW optional	0 heavier than needed	++
Versioned imports, relational history (FR-03)	++	-	+
Deployment effort (NFR-03)	+ one container with extensions	- larger memory footprint	++
Fit with supervisor's document	++ names PostgreSQL FTS and vector indexes	-	-
Kazakh text configuration	0 no Kazakh stemmer in standard configs (to verify); use `simple` config plus trigram	0 analyzers available, not verified	0
Decision: A. With a few hundred rows an exact (unindexed) vector scan is fast, so HNSW is an optimisation to measure, not a requirement (pgvector docs: exact scan is fine for a few thousand rows). Revisit if the registry grows by orders of magnitude (ADR-002).			

5.3 Decision 3: embedding model

Criterion	A. BGE M3	B. multilingual E5 small/base/large	C. Sentence-Transformers multilingual MiniLM family
Kazakh and Russian coverage	+ 100+ languages claimed	+ multilingual pre-training	0 to verify for Kazakh
Model size / CPU latency (NFR-02)	- larger, to measure	++ small and base are light	++ lightest
Extra functions	++ dense, sparse, multi-vector, long input	0 dense only	0 dense only
Licence, hosting	to check	to check	to check
Evidence on this task	none	none	none
Sources: BGE M3 (L21), multilingual E5 (L22), multilingual MiniLM by distillation (L23); benchmark winners differ by task (L24).			
Decision: do not choose by reading. Evaluate A and B (small or base) on the query set of section 9 and keep the cheaper model if the gap is small. Kazakh quality is a stated risk (R3). **Early check (task DIP-38, by M1):** encode the 243 titles and 100 sample queries on the CPU of the planned deployment machine with BGE M3 and multilingual E5 small and base, and record query-encoding latency (p50, p95), start-up time and memory. A model whose single-query encoding alone approaches the 300 ms budget of HS3 is dropped before the relevance comparison. (ADR-003)			

5.4 Decision 4: backend framework

Criterion	A. FastAPI	B. Django + DRF	C. Node.js + Express
Python ML and embedding stack (FR-07)	++ same language	++	- separate service for embeddings
Automatic OpenAPI docs (FR-10, 11)	++ built in	+ add-on	- add-on
Async, performance (NFR-02)	++	0	+
Fit with supervisor's document	++	-	-
Team skill	++ Nurzhan (FastAPI, PostgreSQL experience per justification)	+	++ Abilkhaiyr

Criterion	A. FastAPI	B. Django + DRF	C. Node.js + Express
Decision: A for the API and search. Node and Express stay as the client toolchain and as the data-integration scripts (below) (ADR-004).			

5.5 Decision 5: data integration module

Criterion	A. Node script normalises source to JSON, posts to API	B. Python importer inside the backend
Matches the role split (Frontend & Data Integration)	++	-
Shared validation with the database schema (FR-02)	0 duplicated unless a JSON Schema is shared	++
Parsing of xlsx list tables (irregular layout, section 2.6) and, if needed, PDF or HTML copies	+ libraries exist in both ecosystems; the xlsx of 15 Sep 2026 is in hand	+
Testing and reuse in evaluation	0	++
Decision: A with a versioned JSON contract (`registry-import.schema.json`) that the API validates with Pydantic, so validation lives in one place. Fall back to B if a later edition needs Python-only tooling (ADR-005).		

5.6 Decision 6: frontend

Vue 3 is chosen by the frontend owner. Alternatives considered: server-rendered pages (simpler, fewer interactions, weaker filters and table UX) and React. Decision: Vue 3 with a small component library or plain CSS; table with sorting, filter chips, pagination.

5.7 Summary

Decision	Choice	Alternatives	Main reason
Retrieval strategy	hybrid FTS + trigram + vector, RRF	FTS only, vector only	covers exact, typo and semantic queries
Engine	PostgreSQL + pg_trgm + pgvector	Elasticsearch/OpenSearch, SQLite + FAISS	single system, small data, supervisor's plan
Embedding model	BGE M3 vs multilingual E5, decide by measurement	MiniLM family	Kazakh and Russian quality vs CPU cost
Backend	FastAPI	Django, Express	Python ML stack, OpenAPI
Data integration	Node scripts with shared JSON contract	Python importer	role split, one validation point
Frontend	Vue 3	server-rendered, React	frontend owner's choice

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6. Requirements Specification

Priority: **M** must, **S** should, **C** could (MoSCoW without "won't"; see out of scope in section 1). Status: **Planned** unless stated. Objective ids O1-O6 are from section 1. Items marked (A) rest on assumptions about user needs that the discussions of section 2 must confirm.

6.1 Functional requirements

ID	Requirement	Pri	Obj	Acceptance criterion
FR-01	Import the registry source into normalised journal records through one repeatable command or endpoint	M	O1	Importing the same source twice gives identical data and no duplicates; import log lists counts of added, changed, removed records
FR-02	Validate records: required fields, ISSN format and check digit, duplicates, unknown list level	M	O1	A test file with 20 seeded errors is rejected or flagged for all 20; errors are listed with row numbers
FR-03	Keep each import as a version with date and source reference; show the diff between versions	M	O1	After two imports with 3 known changes, the diff reports exactly those 3; the previous version stays retrievable
FR-04	Store per journal: titles and aliases in kk, ru, en, ISSN, eISSN, publisher, list level, subject areas, languages, website, inclusion order and date, status	M	O2	Schema migration creates all fields; sample record with kk, ru and en titles round-trips unchanged (UTF-8)
FR-05	Full-text search over titles, aliases, publisher and subjects	M	O3	Query of an exact title returns that journal at rank 1 for at least 95% of the evaluation set exact-title queries
FR-06	Fuzzy search tolerant to typos and partial input (trigram similarity)	M	O3	Typo queries with 1-2 edits reach Recall@5 of at least 0.90 (HS2 for the fuzzy component alone is reported, not required)
FR-07	Semantic search using multilingual embeddings for topic and cross-language queries	M	O3	Embeddings stored for every journal; topic queries return relevant journals per the evaluation judgements; cross-language set is reported per language
FR-08	Fuse FTS, fuzzy and vector rankings with Reciprocal Rank Fusion; parameter k and method weights configurable	M	O3	Switching `fusion=rrf
FR-09	ISSN lookup with hyphen and case normalisation	M	O3	`1234–5678`, `12345678`, `1234 5678` return the same journal; invalid ISSN returns a clear message
FR-10	Search API `GET /api/search` with query, filters (level, subject, language, status), method selector, pagination	M	O2	OpenAPI document describes all parameters; contract tests pass
FR-11	Journal API `GET /api/journals/{id}` with status, level, history of changes and source version	M	O2	Response includes every field of FR-04 and history entries for a journal that changed between versions
FR-12	Import API protected by an admin key, with dry-run mode	M	O2	Request without key returns 401; dry-run changes no data
FR-13	Filters in the UI: level, subject, language, status	M	O4	Each filter reduces the result list correctly on a test data set; filters combine
FR-14	Search page with query box, results table, sorting, pagination	M	O4	Typing a query shows results; table sorts by title and level; page size selectable
FR-15	Journal card or detail page with all metadata and change history	M	O4	Open from a result; all FR-04 fields visible
FR-16	Status check: a clear answer "recommended / not found", level, as-of date, source reference, for a title or ISSN (A)	M	O4	For 10 test journals with known status the UI answer matches; "not found" shows search suggestions
FR-17	Input validation, loading, empty and error states in the UI	M	O4	Empty query, too long query, API down and no results each show a specific message, no blank screen
FR-18	Show data version date and source on every page	M	O4	Footer or header shows date of current import
FR-19	Evaluation harness: run a query set through each method (FTS, fuzzy, vector, hybrid variants) and write metrics	M	O5	One command writes Recall@k, MRR, nDCG@10 and latency per method, language and query type to a report file
FR-20	Export results as CSV (C)	C	O4	CSV opens in a spreadsheet with correct Cyrillic text
FR-21	Interface language switch kk / ru / en (A)	S	O4	Labels switch language; search not affected
FR-22	Optional anonymous query log for tuning, off by default	C	O5	When off, nothing stored; when on, only query text and timestamp

6.2 Non-functional requirements

ID	Category	Requirement	Pri	Acceptance criterion
NFR-01	Relevance	HS1 and HS2 targets met on the evaluation set	M	Report in section 9 shows MRR@10 and Recall@5 per method
NFR-02	Performance	p95 search response at most 300 ms on the full registry on the deployment machine, warm cache	M	100 queries, 5 repeats, p95 reported
NFR-03	Deployability	`docker compose up` starts database, API and client from a clean checkout	M	Followed on a second machine by the other team member
NFR-04	Security	No secrets in the repository; admin key from environment; parameterised SQL; inputs length-limited	M	Secret scan clean; test injecting SQL metacharacters returns normal results
NFR-05	Data integrity and provenance	Every journal record links to the import version it came from	M	Query joins journal to version for all rows
NFR-06	Usability	Results visible with at most 2 interactions; layout works on a phone-sized screen; keyboard navigable	S	3 users complete 3 tasks (section 9); viewport check at 375 px
NFR-07	Maintainability	Unit and integration tests run in one command; modules separated (import, search, API, UI)	S	Test command passes in CI or locally; coverage of the search module reported
NFR-08	Reproducibility	Pinned dependencies, fixed evaluation set and random seeds, recorded model version	M	Two evaluation runs give identical metrics for the same data
NFR-09	Privacy	No personal data stored beyond optional query text	M	Review of tables and logs
NFR-10	Localisation	Correct handling of Kazakh letters (ә, ғ, қ, ң, ө, ұ, ү, һ, і), Cyrillic case folding and Latin transliteration	M	Test cases for each letter find the journal under both spellings
NFR-11	Observability	Health endpoint and request logging with timing	S	`/health` returns 200; log lines contain duration
NFR-12	Compatibility	Current Chrome, Firefox, Safari, Edge	S	Smoke test on each

6.3 Traceability to objectives and hypotheses

Objective	Requirements
O1 Registry data	FR-01, FR-02, FR-03, NFR-05
O2 Backend	FR-04, FR-10, FR-11, FR-12, NFR-04, NFR-11
O3 Search core	FR-05 to FR-09, NFR-02, NFR-10
O4 Client	FR-13 to FR-18, FR-20, FR-21, NFR-06, NFR-12
O5 Evaluation	FR-19, FR-22, NFR-01, NFR-02, NFR-08
O6 Deployment and quality	NFR-03, NFR-07, NFR-09
HS1, HS2	FR-19, NFR-01

Objective	Requirements
HS3	NFR-02

6.4 Assumptions and dependencies

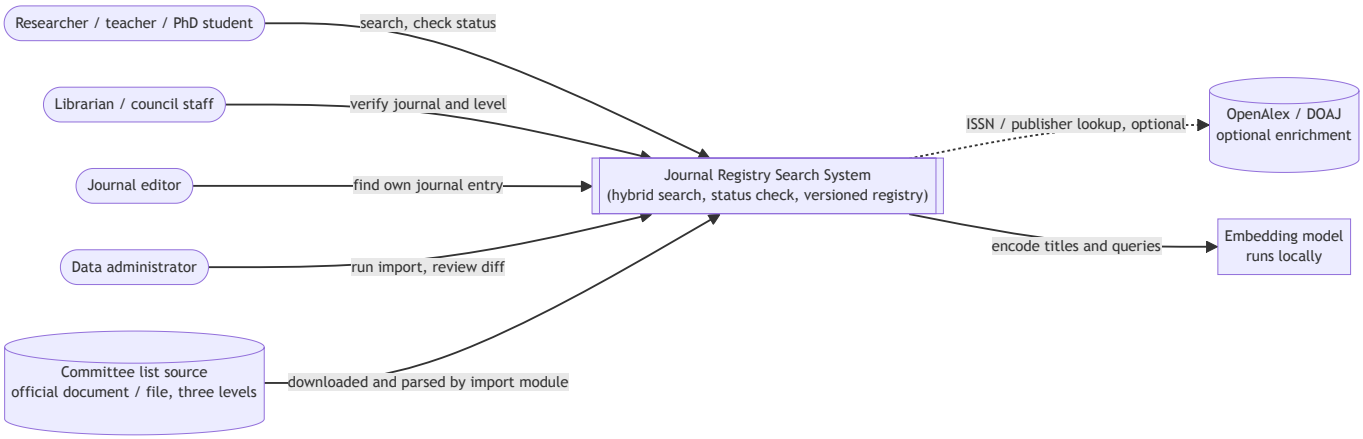
- The registry source can be obtained and parsed (section 10, risk R1).
- List levels and criteria follow the 2025 order; changes are handled by versioned import.
- Thresholds are targets, revised with the supervisor if the first baseline run shows them unrealistic, never after seeing final results.

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7. Architecture and Diagrams

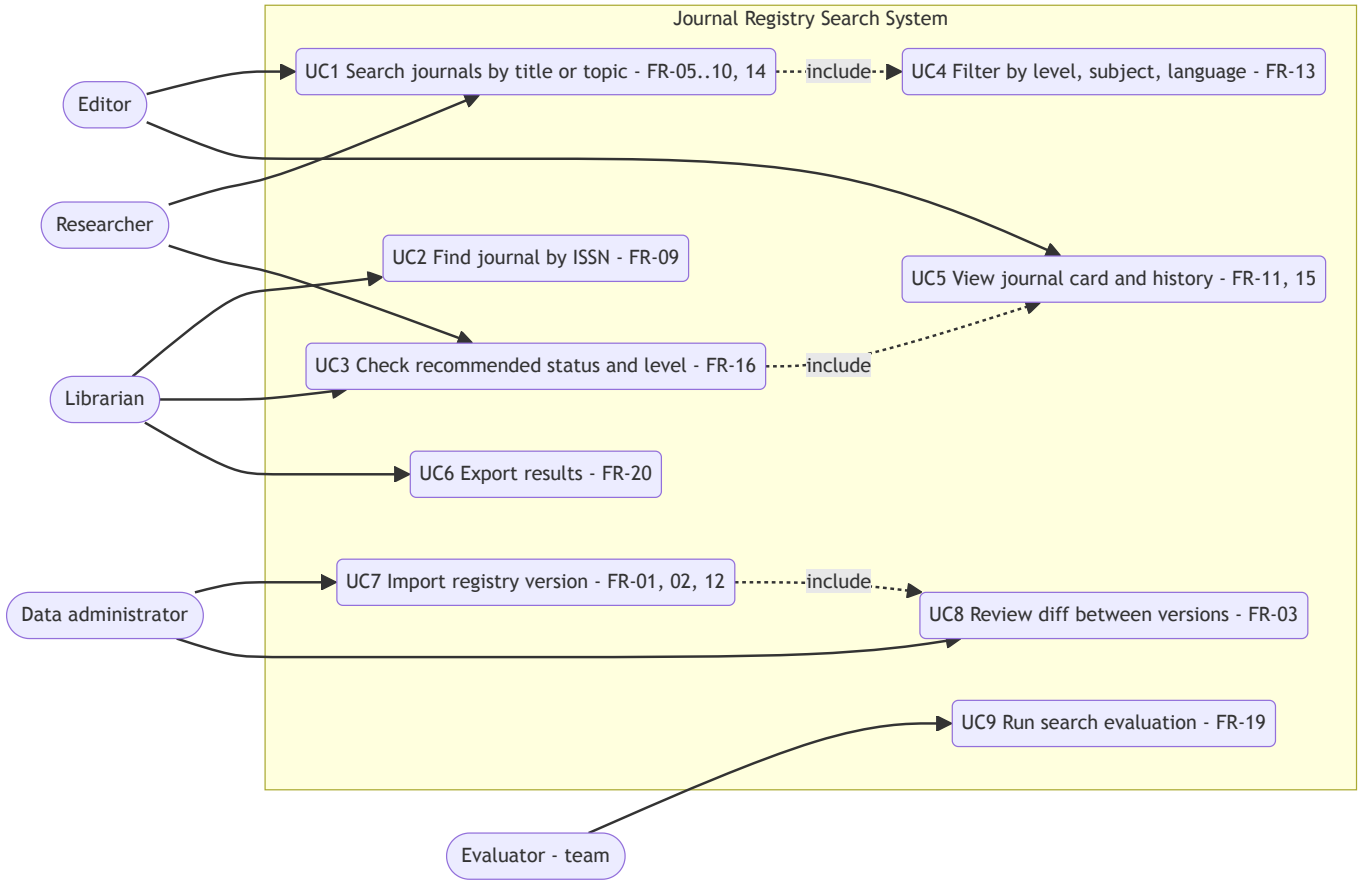
Readable SVG exports are in `diagrams/exports/`. Editable Mermaid sources: `diagrams/context.mmd`, `diagrams/usecase.mmd`, `diagrams/component.mmd`, `diagrams/erd_registry.mermaid`. GitHub renders the code blocks below; the files are the source of truth.

7.1 System context

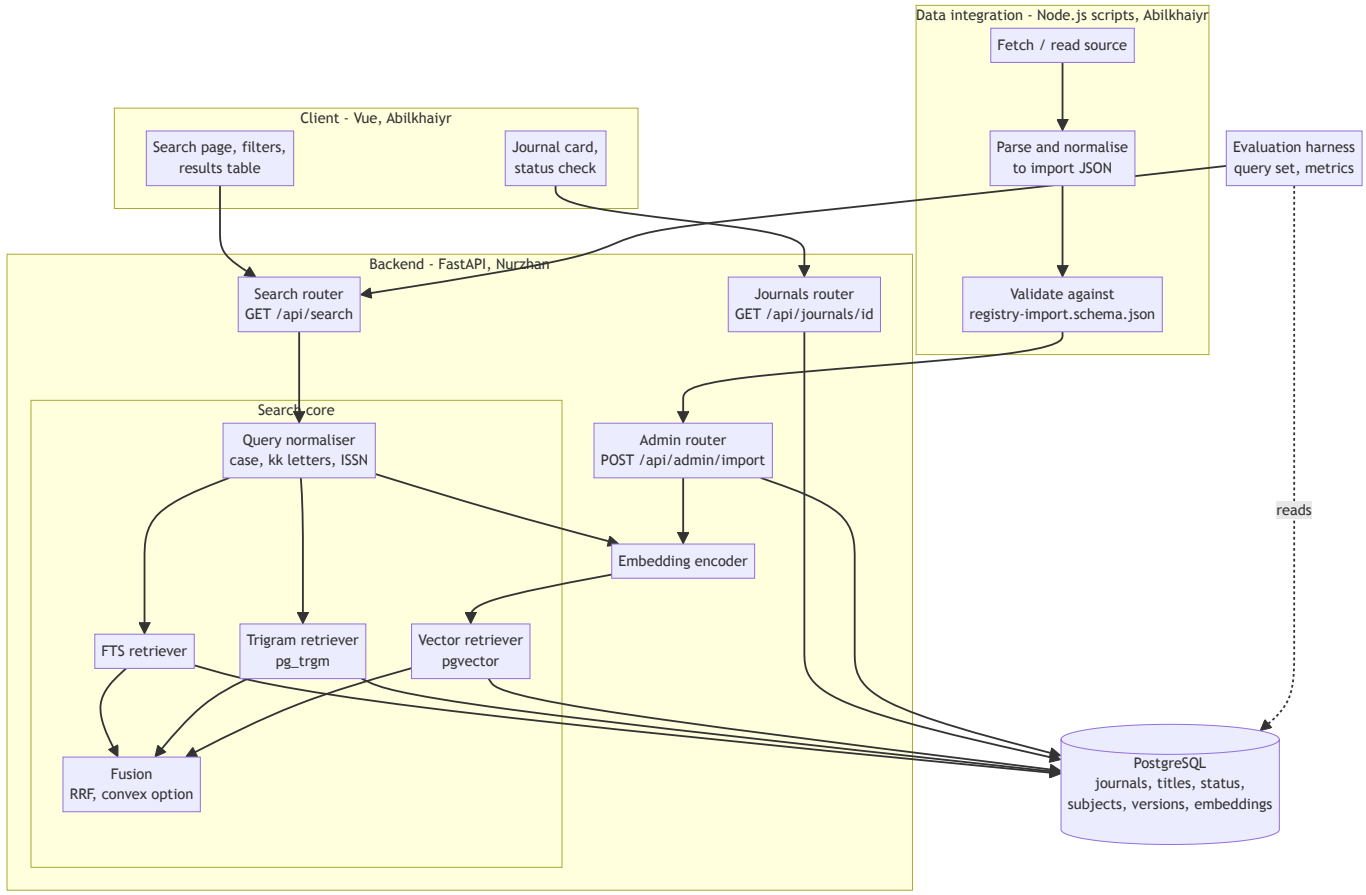


External elements: the Committee list source (an official file; the edition of 15 Sep 2026 is an xlsx, regular publication in this format is still to be confirmed), an embedding model that runs locally, and optional public APIs (OpenAlex, DOAJ) for enrichment. Users do not log in; only the data administrator uses a protected import endpoint.

7.2 Use cases

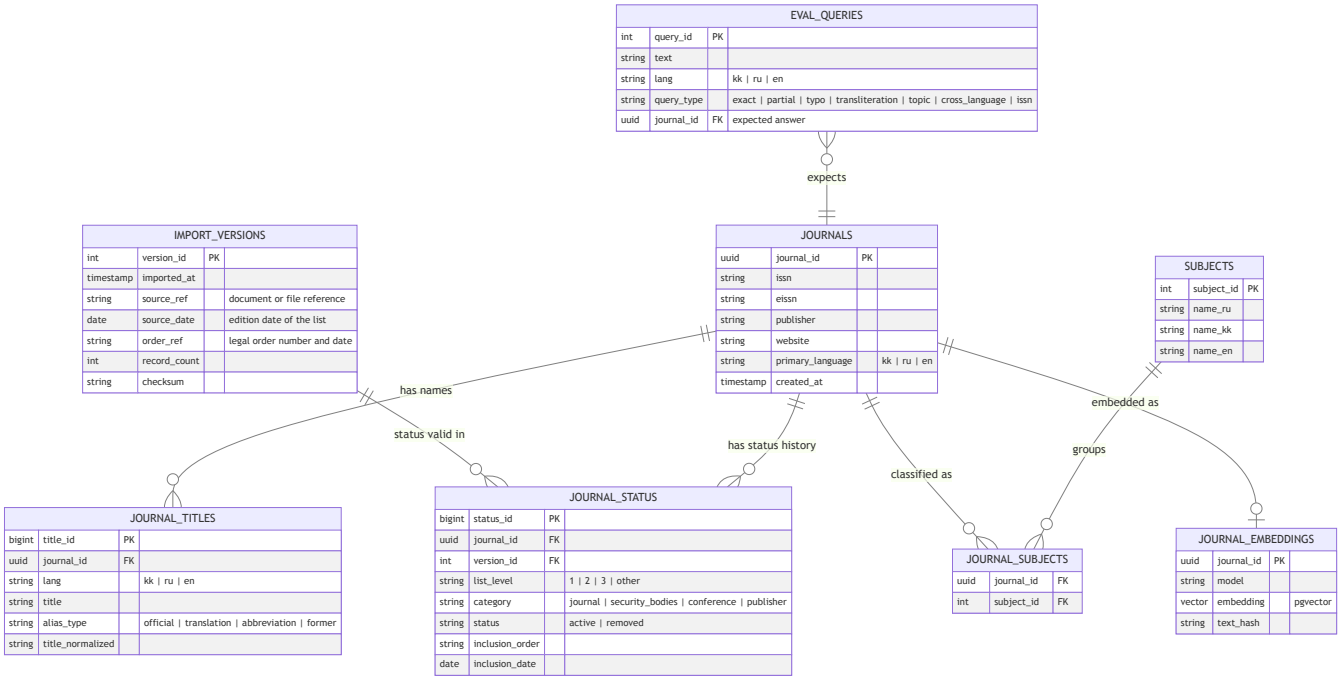


7.3 Component diagram (draft)



Component	Responsibility	Owner	Technology	Requirements
Vue frontend	search page, filters, results table, journal card, status check, validation and error states	Abilkhaiyr	Vue, HTML/CSS	FR-13 to 18, 20, 21
Data integration	read the official source, parse, normalise to import JSON, validate, call import API, show diff summary	Abilkhaiyr	Node.js scripts, JSON Schema	FR-01 to 03
Import router	accept import JSON, dry-run, write version and records, trigger embedding	Nurzhan	FastAPI, Pydantic	FR-01, 03, 12
Query normaliser	case folding, Kazakh letters, ISSN normalisation, trimming	Nurzhan	Python	FR-09, NFR-10
FTS retriever	full-text ranking over titles, aliases, publisher, subjects	Nurzhan	PostgreSQL FTS (`tsvector`, GIN)	FR-05
Trigram retriever	typo-tolerant title matching	Nurzhan	`pg_trgm`, GIN index	FR-06
Vector retriever	semantic neighbours of the query embedding	Nurzhan	`pgvector`, exact scan or HNSW	FR-07
Fusion	combine three ranked lists, RRF (k = 60) and convex option	Nurzhan	Python	FR-08
Search and journal routers	public API, filters, pagination, OpenAPI	Nurzhan	FastAPI	FR-10, 11
Evaluation harness	run query set per method, compute metrics and latency	Nurzhan with Abilkhaiyr's query set	Python	FR-19
Deployment	database, API and client in containers	Nurzhan	Docker Compose	NFR-03

7.4 Data model



The ERD is a design; no registry schema exists in code yet. Journals are separated from their status per import version, so a journal can change level or be removed without losing history. Titles live in their own table, one row per language and alias, which supports cross-language matching and abbreviations. Evaluation queries are stored with the expected journal.

7.5 Search design

1. **Normalise** the query: Unicode NFKC, lower case, Kazakh and Russian letter variants, remove punctuation, detect ISSN pattern (an ISSN query bypasses fusion and does an exact lookup).
2. **Retrieve** three ranked lists (top 50 each): FTS over titles and aliases (`simple` configuration plus language-specific variants where available), trigram similarity over normalised titles, vector similarity between query embedding and journal embedding (built from titles in all languages plus subject names).
3. **Fuse** with RRF: score(d) = sum over lists of 1 / (k + rank_list(d)), k = 60 by default. Ablation: convex combination of normalised scores with one weight tuned on a development split of the query set.
4. **Return** top N with the matching method per result, filters applied before ranking.
5. **Index choices:** GIN on `tsvector` and on `gin_trgm_ops`; vector column with exact scan first, HNSW added only if measurements show a need (registry has hundreds of rows).

7.6 API sketch

Endpoint	Purpose
`GET /api/search?q=&level=&subject=&lang=&status=&method=hybrid&page=`	ranked journals, `method` in `fts
`GET /api/journals/{id}`	full record with status history
`GET /api/issn/{issn}`	exact lookup
`GET /api/status?q=`	status-check answer: recommended yes/no, level, as-of date, source
`POST /api/admin/import?dry_run=`	import JSON (admin key)
`GET /api/versions`, `GET /api/versions/{a}/diff/{b}`	import history and diff
`GET /health`	health check
Examples with JSON are in Appendix E.	

7.7 Traceability: requirements, components, data, scenarios

Requirement	Component	Data	Scenario (section 8)
FR-01, 02, 03, 12	Data integration, import router	`IMPORT_VERSIONS`, `JOURNALS`, `JOURNAL_STATUS`	S4
FR-04	schema	all registry tables	all
FR-05, 06, 07, 08, 09	search core	titles, embeddings	S1, S2
FR-10, 11	routers	journals, status	S1, S3
FR-13, 14, 15	client	via API	S1, S2
FR-16, 18	client, status endpoint	`JOURNAL_STATUS`, `IMPORT_VERSIONS`	S3
FR-19	evaluation harness	`EVAL_QUERIES`	S5
NFR-02	search core, indexes	all	S5
NFR-03	deployment	n/a	all

7.8 Main flows

1. **Import:** administrator runs the integration script; it parses the source, validates against the JSON contract, and posts it in dry-run mode, then for real. The API stores a new version, updates journals and status rows, computes embeddings for new or changed titles, and returns a diff summary.
2. **Search:** client calls the search endpoint; the search core normalises the query, runs three retrievers, fuses, applies filters, returns results with the method that found each.
3. **Status check:** client calls the status endpoint; the answer comes from the latest version, with the as-of date and source reference. A "not found" answer comes with near matches.

4. **Evaluation:** harness replays the query set with each `method` value, records ranks and latency, writes the report.

7.9 Constraints and risks in the design

- The source layout is irregular (merged headers, mixed-language cells, free-text dates, section 2.6) and may change between editions, so the integration module is the least certain component.
- With hundreds of rows, exact search can be near-perfect for title queries; the value of the vector component must be shown on topic and cross-language queries.
- Embedding models consume memory and start-up time; the container must load the model once.
- Kazakh full-text configuration is limited; the trigram retriever is the safety net.

---

8. UX and Prototype

Nothing of the core interface exists yet. This section shows **mid-fidelity HTML mockups (sources in `diagrams/mockups/`, PNG exports in `diagrams/exports/`) and API examples** for the key scenarios. Journal names, ISSNs and levels in the mockups are **placeholders**, not real registry entries (the registry has not been imported yet). Real registry rows replace them when import (DIP-11, DIP-13) is done.

8.1 Scenarios

ID	Actor	Goal	Steps	Requirements
S1	Researcher	find a journal from an imperfect title	type a title with a typo or in another language, see ranked results with the match reason, open the journal card	FR-05 to 08, 14, 15
S2	Researcher	find journals on a topic	type a topic in Russian or English, filter by level, compare candidates	FR-07, 13, 14
S3	Librarian	confirm whether a journal is recommended and at which level	enter title or ISSN on the status page, read the answer with as-of date and source, optionally export	FR-09, 16, 18, 20
S4	Data administrator	update the registry	run the import script in dry-run mode, review the diff, confirm the import	FR-01 to 03, 12
S5	Team (evaluator)	measure search quality	run the harness, read the metrics report	FR-19

8.2 Mockup A: search page (S1, S2)

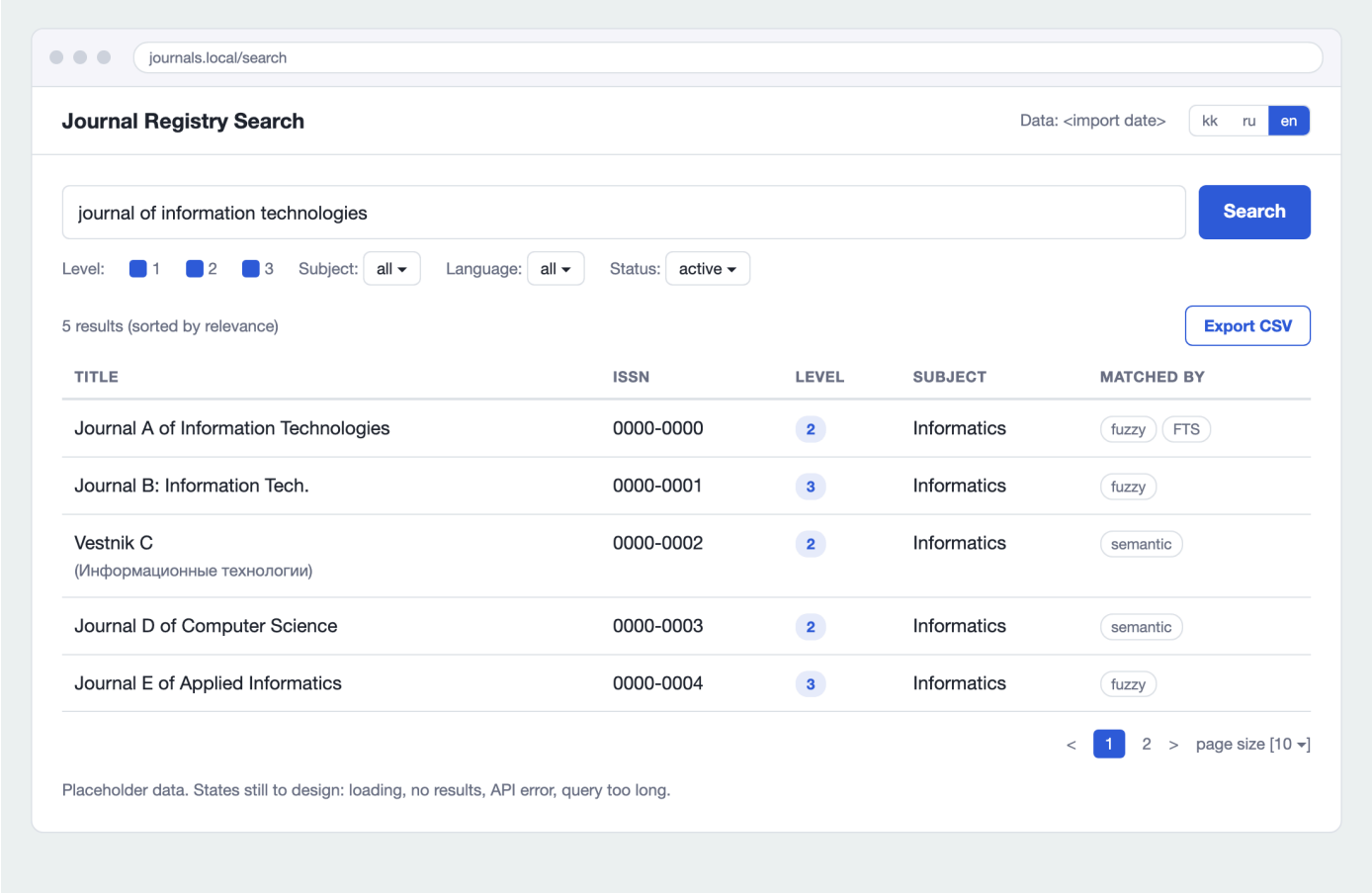


Figure 8.1. Mockup A: search page (S1, S2), placeholder data. States to design: loading spinner, "no results, try fewer words" with suggestions, error banner when the API is unreachable, too-long query message.

8.3 Mockup B: journal card (S1, S3)

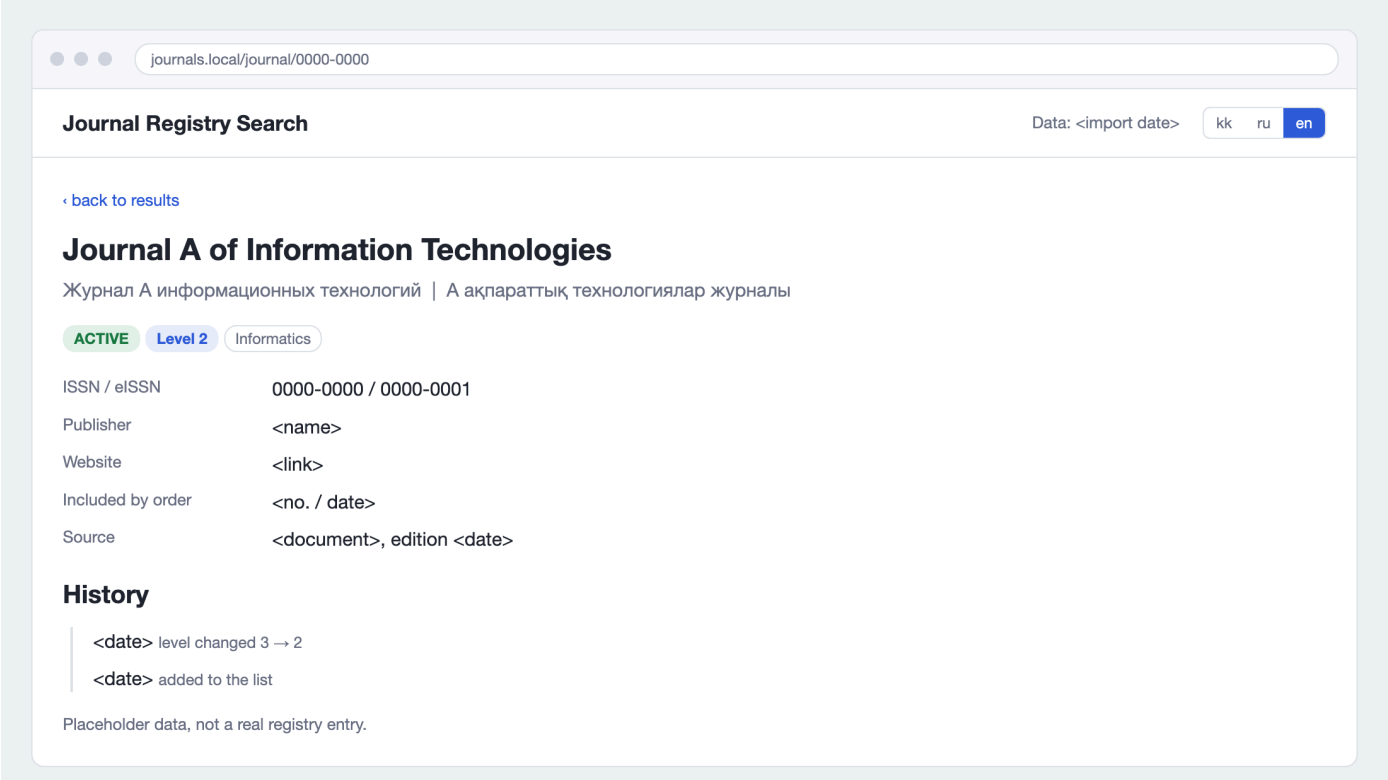


Figure 8.2. Mockup B: journal card (S1, S3), placeholder data.

8.4 Mockup C: status check (S3)

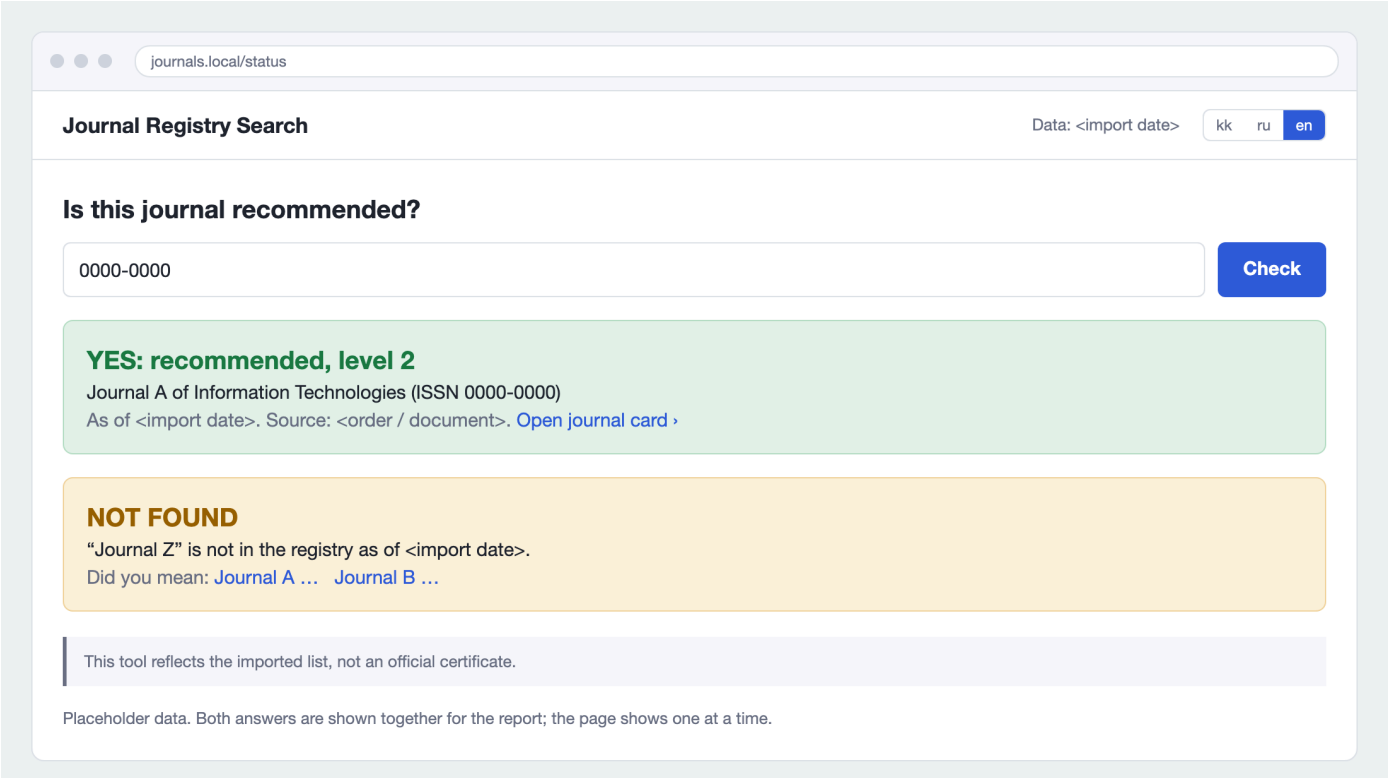


Figure 8.3. Mockup C: status check (S3), placeholder data; the YES and NOT FOUND answers are shown together, the page shows one at a time. The disclaimer matters: the system is a convenience tool, not an official decision.

8.5 Mockup D: import review (S4), CLI or admin page

```
$ node integration/import.js --source <file> --dry-run
Parsed 243 rows, 0 invalid
Version diff vs previous (<date>):
  added   3   removed 2   changed 5 (level: 3, subject: 1, website: 1)
  warnings: 1 duplicate ISSN (row 118)
Dry run only. Re-run with --apply to import.
```

Figure 8.4. Mockup D: import dry-run output (S4), numbers illustrative.

8.6 API examples

Requests and responses for search, status check, journal detail and import dry-run are in Appendix E (shapes only, with placeholder data).

8.7 Accessibility and responsiveness

Layout works at 375 px width (table becomes cards), keyboard-reachable controls, visible focus, contrast checked, text size user-adjustable, labels in kk, ru and en.

journals.local/search

Journal Registry Search

kk ru en

journal of information technologies

Search

Level: 1 2 3 Subject: all

Language: all Status: active

5 results (sorted by relevance) Export CSV

Journal A of Information Technologies

0000-0000

2

Informatics

fuzzy FTS

Journal B: Information Tech.

0000-0001

3

Informatics

fuzzy

Vestnik C

(Информационные технологии)



Figure 8.5. Mockup A at 375 px width: the table becomes cards.

8.8 Open questions for the user discussions (section 2)

Which columns users want in the table; whether the status page should also state the criteria of each level; whether international auto-included journals must be searchable; whether users want alerts when a journal leaves the list.

---

9. Verification and Evaluation Plan

Goal: show that the core system meets its requirements (section 6) and answers the research question: does hybrid retrieval find the intended journal more reliably than each method alone on realistic trilingual queries, within the latency target. Requirement ids refer to section 6.

9.1 Test levels

Level	What	Tools (to be fixed in DIP tasks)	Requirements
Unit	query normaliser (Kazakh letters, case, ISSN), ISSN checksum, RRF and convex fusion functions, import validation rules, diff computation	Python `pytest`; JavaScript unit tests for parsers	FR-02, 03, 08, 09, NFR-10
Integration	import file to database to search result, version diff, embedding step	`pytest` with a test PostgreSQL container	FR-01, 03, 04, 05-07
API contract	every endpoint against its OpenAPI description; 401 without admin key; dry-run changes nothing	contract tests generated from the OpenAPI file	FR-10, 11, 12, NFR-04
Frontend component and end-to-end	filters, sorting, pagination, empty, loading and error states, status check on 10 known journals	component tests and a browser end-to-end tool (Playwright or Cypress)	FR-13 to 18
Search quality	metrics on the evaluation set (9.2-9.5)	evaluation harness (FR-19)	NFR-01
Performance	latency under repeated and concurrent queries	script replaying queries, 100 queries x 5 repeats, concurrency 1 and 10	NFR-02
Security and hygiene	secret scan, SQL metacharacter queries, over-long input	scripted checks	NFR-04
Usability and accessibility	3-5 users, 3 tasks; keyboard-only pass; 375 px viewport	task sheet, timer, short questionnaire	NFR-06
Deployment	clean checkout on a second machine, `docker compose up`, smoke test	checklist	NFR-03

9.2 Evaluation set (queries and ground truth)

Corpus: the imported registry (243 rows in the edition of 15 Sep 2026; IT/CS subset of 15 journals used for topic and subject queries). Frozen with a version hash before evaluation.

Query types and planned sizes (at least 400 queries in total; sizes to adjust to the registry):

Type	How made	Planned	Ground truth
Exact title	official titles in each language	60	the journal
Partial title	first or last words, dropped words	60	the journal
Typo	1-2 random character edits, fixed seed	100	the journal
Transliteration	Cyrillic to Latin by a fixed scheme, and Latin to Cyrillic	40	the journal
ISSN	with and without hyphen and spaces	30	the journal
Cross-language	title in another language than the query	50	the journal
Topic	written by team and by survey respondents, in kk, ru, en	60	set of relevant journals judged by two annotators (pooled top 10 of all methods)

Avoiding circularity. Queries generated from the titles favour lexical methods. Therefore (a) real user queries are collected (survey Q13 and discussions U1-U5, section 2.4; target at least 50 queries in total), reported as a separate subset, and (b) topic and cross-language queries are written by people, not generated.

Judgments. Two annotators judge topic queries independently on a 0/1/2 scale; agreement (Cohen's kappa) reported; disagreements resolved by discussion with the supervisor.

Split. Queries are split by journal into a development part (30%) for tuning RRF k, convex weight and trigram threshold, and a test part (70%) used once per final configuration.

9.3 Systems compared

ID	System	Role
B0	substring match (`ILIKE`) on titles	stands for document search or Ctrl+F, today's practice
M1	FTS only	lexical baseline
M2	trigram only	fuzzy baseline
M3	vector only, embedding model A	semantic baseline
M4	vector only, embedding model B	model comparison (BGE M3 vs multilingual E5)
H1	hybrid RRF, FTS + trigram + vector (best model)	proposed system
H2	hybrid convex combination, weight tuned on development queries	ablation
H3	hybrid RRF without trigram	ablation
H4	hybrid RRF without vector	ablation, shows what semantic search adds
Sensitivity: RRF k in {10, 30, 60, 100}.		

9.4 Metrics and success criteria

Metric	Definition	Success criterion
Recall@1, @5, @10	share of queries whose target is within top k	HS2: H1 Recall@5 >= 0.95 overall and >= 0.90 on typo queries
MRR@10	mean reciprocal rank of the target	HS1: H1 at least 0.05 above the best of M1-M4 on the whole test set
nDCG@10	for topic queries with graded judgments	report; H1 not below the best single method
Per language, per query type	same metrics split by kk, ru, en and by type	reported; flag any cell more than 0.10 below the mean
Latency p50, p95	API response time, warm cache, 100 queries x 5 repeats	HS3: p95 <= 300 ms for H1 on the deployment machine
Status-check accuracy	correct answer on 30 journals with known status (including removed ones and non-members)	100% on members, at least 95% overall; wrong answers are reviewed one by one

Metric	Definition	Success criterion
Decision rule: a hypothesis is "supported" when the criterion holds on the test part and the paired-bootstrap 95% confidence interval of the difference excludes 0. If a criterion fails, the result is reported as measured; thresholds are not changed after seeing test results.		

9.5 Measurement conditions

- Hardware and software: record CPU, RAM, whether a GPU is used for embedding, PostgreSQL, pgvector and library versions; same container image for all runs.
- Warm cache and cold start measured separately; embedding model loaded once.
- Same normalisation and same candidate cut-off (top 50) for all methods.
- Fixed random seeds for typo generation and any sampling; evaluation set and registry version hashed and logged with every result.
- Paired bootstrap over queries (1,000 resamples) for differences between methods; no multiple-testing claims beyond the pre-listed comparisons.

9.6 Prototype acceptance checks

Check	Procedure	Pass
Import	import two versions with 3 known changes	diff shows exactly 3; second import creates no duplicates
Status check	30 journals with known status	answers as in 9.4
Search smoke	10 predefined queries (kk, ru, en, typo, ISSN, topic)	expected journal in top 5 for at least 9
UI states	empty query, long query, API down, no results	each shows a specific message
Deployment	second machine, clean checkout	`docker compose up` gives working system
Usability	3 users x 3 tasks (find by typo, check status, filter by level)	at least 80% completed, median under 2 minutes

9.7 Schedule (aligned with the supervisor's plan)

When	Activity
Oct-Nov 2026	obtain and inspect the source; draft query-set method; run the user survey and collect real queries; **early checks by M1 (30 Nov): HS1 pilot and CPU latency of the embedding models (below)**
Dec 2026 - Jan 2027	unit tests for normaliser, validation, fusion; first baselines M1-M3 on a prototype set
Feb-Mar 2027	integration and API tests; full query set; embedding model comparison; tuning on development queries
Apr-May 2027	final test-part run, latency tests, usability sessions, deployment check, thesis write-up

Early checks before the full evaluation (by M1, 30 Nov 2026).

1. *HS1 feasibility pilot (task DIP-37).* With a small prototype (FTS, trigram and one embedding model over the 243 imported rows) and about 100 queries of all types written before the run, measure MRR@10 of M1, M2, M3 and H1 on the development part only. Purpose: see whether the best single method already leaves less than 0.05 of headroom (ceiling effect on a small registry). If so, the team proposes to the supervisor, before any test-part run, a revised wording of HS1 (for example, the margin measured on typo, cross-language and topic queries only, or a non-inferiority claim with a latency or robustness advantage). The test part is not touched and no threshold is changed afterwards, as required by section 6.4.
2. *Embedding latency on CPU (task DIP-38).* Measured as described in section 5.3; the numbers go into the risk R8 and into the choice of the models that enter the full comparison.

9.8 Threats to validity

- Registry is small; high scores on title queries are expected from all methods, so the value of hybrid search is judged mostly on typo, cross-language and topic queries. Whether the HS1 margin is reachable at all is checked early (pilot above).
- Generated queries are not real user behaviour (mitigated by survey queries and human-written queries).
- Embedding quality for Kazakh may be low (expected, to be measured); results are reported per language.
- The source may change between evaluation and defence; the registry version is frozen for evaluation.
- Judgments for topic queries are subjective; agreement is reported.

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10. Project Management and Risks

10.1 Repository and tracker

Item	Value
Git repository	https://github.com/nurzhqn0/indexing-system (branch `main`, tag `pre-defense-1`; contribution status in 10.8)
YouTrack project	https://indexing-system.youtrack.cloud/dashboard?id=177-0, key `DIP`, tasks in Appendix D
Documentation	this report, `docs/REPORT.md`; README in the repository root
Access	supervisor (Symbat Nurgaliyeva), Abilkhaiyr Sarsenbay and the committee to be added as collaborators before the pre-defense

10.2 Roles and supervision

Roles are those stated in the supervisor's justification.

Area	Nurzhan Izimbetov (Backend & Search Core)	Abilkhaiyr Sarsenbay (Frontend & Data Integration)
Server architecture, schema, FastAPI	owner	reviewer
Search core (FTS, fuzzy, vector, RRF), performance	owner	reviewer
Containerisation and deployment	owner	reviewer
Vue frontend, user scenarios, validation, error handling	reviewer	owner
Registry data preparation and synchronisation	reviewer (schema, import API)	owner
API contract (OpenAPI, import JSON schema)	co-owner	co-owner
Evaluation: harness and metrics	owner	reviewer
Evaluation: query set and judgments	co-owner	co-owner
Requirements, report, thesis, defence	shared	shared
Each student commits from their own GitHub account. Each change gets a review from the other student; the supervisor reviews significant changes at agreed checkpoints.		
Supervision (from the justification): weekly meetings with both students, in person or online, with code review, integration checks and schedule control; consultations on architecture and search methods; intermediate demonstrations; agreement of requirements for the final product. If work falls behind, scope is reduced while key results are kept. Meeting dates are to be logged in section 11.		

10.3 Milestones

Phases follow the supervisor's plan; dates inside phases are targets.

ID	Milestone	Target	Exit criterion
M0	Pre-defense 1: analysis and design package	24 Oct 2026	all 11 sections in this report, repository with README
M1	Analysis finished, data and design ready	30 Nov 2026	requirements agreed with supervisor, schema and import contract defined, official source obtained and inspected, high-fidelity mockups, backend and client skeletons in containers, HS1 pilot (DIP-37) and CPU latency of the embedding models (DIP-38) reported
M2	Search core and client components	31 Jan 2027	import, FTS, trigram, vector, fusion working through the API on real registry data; search page, table and filters against the API
M3	Integration	31 Mar 2027	client and server integrated, status check, error handling, integration and end-to-end tests, evaluation harness and full query set
M4	Deployment, evaluation, thesis	20 May 2027	deployed system, final evaluation, usability test, written thesis with individual sections, defence materials

10.4 Tasks and outcomes

Task table in Appendix D: id, title, owner, expected outcome, acceptance criterion, priority, dependency. No task claims completed work that does not exist.

10.5 Risks

Scale: L low, M medium, H high.

ID	Risk	Evidence	Likelihood	Impact	Mitigation	Owner
R1	Official list is not machine-readable, is hard to obtain, or changes format	an xlsx edition of 15 Sep 2026 is in hand, but its layout is irregular (section 2.6)	H	M	obtain each new edition from the Committee, Adilet or prg.kz, build a table parser with manual CSV fallback, versioned import, ask supervisor for official file	Abilkhaiyr
R2	Registry is small, so every method scores high and differences are not significant; the HS1 margin (+0.05 MRR@10) may be unreachable	243 rows in the current file, 15 in the IT/CS subset	H	M	HS1 pilot on the development part by M1 (section 9.7, DIP-37); evaluate by query type (typo, cross-language, topic), include real user queries from the survey, report confidence intervals; optionally add international lists if legally usable	Nurzhan
R3	Weak Kazakh support in FTS and embeddings	no Kazakh stemmer in standard PostgreSQL configurations (to verify); embedding quality for Kazakh unmeasured	M	M	trigram retriever, transliteration table, compare embedding models, report per language	Nurzhan
R4	Unclear whether international auto-included journals belong in scope	List 1 includes WoS and Scopus journals automatically	M	M	ask supervisor (open scope questions, section 1); implement a rule note and an extension point	both
R5	Data licensing or reuse limits of third-party lists	Scopus and WoS lists are vendor data	L	M	use only the official Committee list as core data; public APIs only for enrichment	both
R6	Integration friction between Vue frontend and FastAPI	two owners, two stacks	M	M	contract-first (OpenAPI and JSON schema), mock API for the client, integration check at each weekly meeting	both
R7	Schedule slip in one role blocks the other	client depends on API; API tests depend on data	M	M	mock API and sample data from the first weeks; weekly meetings; supervisor checks individual stages at each meeting	both, supervisor
R8	Embedding model too heavy for the deployment machine, query encoding alone eats the 300 ms budget (HS3)	BGE M3 larger than small E5; not measured yet	M	H	measure CPU latency, start-up time and memory of BGE M3 and E5 small and base by M1 (DIP-38, section 5.3); choose a small model if relevance is similar; cache query embeddings	Nurzhan

ID	Risk	Evidence	Likelihood	Impact	Mitigation	Owner
R9	Evaluation biased by synthetic queries	queries derived from titles favour lexical methods	M	M	real and human-written queries, separate reporting, two-annotator judgments	both
R10	Unknown hosting for final deployment	not discussed	M	M	Docker Compose, ask university IT for a server early, fallback to a cloud VM	Nurzhan
R11	Official list or criteria change during the project	quarterly updates reported	M	L	versioned import; freeze registry version for evaluation	Abilkhaiyr
R13	Thesis writing compressed at the end	final phase is April-May	M	M	write sections as milestones close; this report is reused	both
R14	Low survey response or no volunteers for discussions	survey runs 24 Oct to 6 Nov	M	M	send to several groups through supervisor and department chats, one reminder, lower target to 20 responses and say so in section 2	both

10.6 Dependencies

Source file (R1) gates the import and every dataset-based task. API contract gates the client integration. Schema gates search retrievers. Query set and judgments gate the final evaluation. Server access gates deployment. Supervisor review gates milestones.

10.7 Git and review workflow (course requirement)

- Work is committed straight to `main`; each student commits from their own account.
- Commit message: `DIP-<n>: <what changed>`. The commit body gives the YouTrack link and how the change was checked.
- Tags `pre-defense-1`, `pre-defense-2`, `final-pre-defense`, never moved.
- Commits are made when work is done, not backdated. Earlier work (this report, diagrams) is committed in logical units on the first push.
- Secrets and data: `.gitignore` excludes `.env` and generated data. Configuration examples use placeholders. Personal data (grades, personal circumstances) from administrative documents is not stored in the repository.
- Repository layout (planned): `backend/`, `frontend/`, `integration/`, `evaluation/`, `docs/`, `diagrams/`, `deploy/`.

10.8 Evidence status for the repository requirement

As of 24 Oct 2026 (`git shortlog -sn` on `origin/main`):

Author	Commits	Content
Nurzhan Izimbetov	8	repository scaffold (DIP-2), diagrams and mockups, bibliography and list source file, this report (DIP-1), README
Abilkhaiyr Sarsenbay	3	OJS OAI-PMH enrichment module for the IT/CS subset, `integration/enrichment/` (DIP-36)

Tag `pre-defense-1` marks the package submitted on 24 Oct 2026; this revised report is committed after the tag and is not part of it. Open points: (1) the first commits of Nurzhan carry a placeholder e-mail address, so GitHub does not attribute them to his account; the address is fixed for all new commits and `git shortlog` still shows the earlier ones under his name; (2) Abilkhaiyr's later work on the enrichment module and the data-integration tasks (DIP-9, DIP-11) must be committed from his account as it is done.

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11. Supervisor Feedback

Supervisor: Symbat Nurgaliyeva, PhD, assistant professor.

11.1 Deliverables to review

Item	Section	Decision requested
Proposal, hypotheses, open scope questions	1	answer the open scope questions (list sections in scope, international journals); confirm HS1-HS3
Relevance evidence and survey plan	2	approve survey questions; confirm supervisor as client proxy
Literature review	3	check sources and gaps
Existing systems and technology choices	4, 5	confirm criteria and decisions
Requirements and thresholds (HS1-HS3)	6, 9	confirm priorities and thresholds
Architecture and diagrams	7	check consistency with the justification
Mockups	8	confirm scenarios
Management, roles, risks, tasks	10	confirm plan

11.2 Supervisor review of pre-defense 1 (24 October 2026)

Reviewer: Symbat Nurgaliyeva, PhD, assistant professor. Form: written review of the pre-defense package, reproduced here in English.

General assessment. The pre-defense materials are complete: all 11 sections and Appendices A-E are present. The topic is relevant: today the status of a journal on the Committee's list can only be checked in the official file, and there is no convenient search tool. The students have formulated a testable research question on whether hybrid retrieval (full-text search, trigram matching and vector search, fused by RRF) works better on trilingual queries.

Strengths of the work

- analysis of the real list file (243 entries, an IT/CS subset of 15 journals);
- a clear separation of verified facts from assumptions;
- a well-designed evaluation method: baselines, ablations, development and test split, bootstrap;
- requirements traceability, a risk register and a task plan.

Roles. The roles are well balanced and match the project justification.

Conclusion. The work is recommended for admission to the next stage.

11.3 Feedback log

Date	Channel	Comment (verbatim or close summary)	Section	Decision (accept / modify / reject, why)	Change made	Commit
24 Oct 2026	written review of the pre-defense package	package complete, topic relevant, research question testable; strengths: real list file analysis, verified vs assumed, evaluation design, traceability; roles balanced; recommended for admission to the next stage (full text in 11.2)	all	accept	report revised for milestone M1 (24 Oct 2026 version)	see `git log` after the revision commit

11.4 How feedback is handled

Each comment becomes a YouTrack task or a note on an existing task, gets a decision with reason, and is linked to the commit that implements it. Rejected comments keep their reason in the log.

Appendix A. Literature Search Log

A.1 Structured database search

Concepts. (1) Academic journals and indexing: academic journal, scholarly journal, citation index, digital library, repository, OJS. (2) Extraction and citation parsing: metadata extraction, reference parsing, citation extraction, citation linking, author disambiguation, affiliation normalisation, document layout analysis. (3) Methods: deep learning, NLP, GROBID, LayoutLM, LLM, transformer, BERT, CRF, graph neural networks. (4) Multilingual and regional scope: multilingual, Cyrillic, Kazakh, Russian, cross-lingual, Central Asia, national citation index.

ID	Database	Focus	Filters	Hits	Selected
S1	Scopus	metadata extraction and AI for scholarly material	article, open access, computer science / engineering	156	3
S2	Scopus	multilingual and Cyrillic parsing	article, open access, all subjects	11	2
S3	Scopus	reference parsing, GROBID pipelines	article, open access, computer science	96	2
S4	IEEE Xplore	neural extraction, BERT	journals, open access	2	1
S5	ACM Digital Library	citation graphs, digital libraries	journal articles, open access	205	1
S6	Scopus	regional and national systems, OJS	article, open access	3,130	1
S7	Google Scholar	snowballing on GROBID	peer-reviewed, open-access PDF	363	1
Total				3,963	11

Example Boolean string (S1): `TITLE-ABS-KEY(("academic journal" OR "scholarly" OR "citation index" OR "digital library") AND ("metadata extraction" OR "citation extraction" OR "indexing") AND ("deep learning" OR "machine learning" OR "NLP"))`.

Inclusion criteria. IC1 peer-reviewed journal article; IC2 open access full text; IC3 direct focus on metadata extraction, reference parsing, citation linking, author disambiguation or indexing systems; IC4 quantitative results (precision, recall, F1, error rates) with sample sizes; IC5 English, or explicitly multilingual, Cyrillic or non-Latin corpora.

Exclusion criteria. EC1 editorials, short abstracts, drafts, blog posts, unvetted preprints; EC2 paywalled; EC3 duplicates and superseded workshop drafts; EC4 generic OCR or scraping without layout-aware bibliographic structure; EC5 theory or software announcements without quantitative validation.

A.2 Targeted searches (October 2026)

Web searches for specific targets; bibliographic data checked against publisher, ACL Anthology, arXiv or DOI records.

ID	Query intent	Result used
S8	named baseline parsers	Lopez 2009 (GROBID, L12); Tkaczyk et al. 2015 (CERMINE, L13)
S9	coverage of Scopus, Web of Science and open databases	Visser et al. 2021 (L14); Martín-Martín et al. 2021 (L15)
S10	open metadata source for journals	Priem et al. 2022 (OpenAlex, L16)
S11	national citation index and journal selection in a neighbouring country	Moskaleva et al. 2018 (L17)
S12	coverage of post-Soviet and Central Asian journals in Scopus and Web of Science	Alimova & Brumshteyn 2020, doi 10.3897/ese.2020.e53192 (L18; authors and title checked on the journal page)
S13	rank fusion for hybrid retrieval	Cormack et al. 2009 (L19, DOI checked); Bruch et al. 2022 (L20, arXiv page checked)
S14	multilingual embedding models and benchmarks	Chen et al. 2024 (L21, ACL Anthology); Wang et al. 2024 (L22, arXiv); Reimers & Gurevych 2020 (L23, ACL Anthology); Muennighoff et al. 2023 (L24, ACL Anthology)
S15	Kazakh retrieval and language resources	Yeshpanov et al. 2024 (L25, ACL Anthology); Yeshpanov et al. 2022 (L26, ACL Anthology)
C1	official list of recommended journals and its structure	zakon.kz (order of 3 June 2025), a-articles.kz (list levels and counts), prg.kz and Adilet (list document), Committee presentation (earlier counts)
C9	PostgreSQL vector and trigram support	pgvector docs, PostgreSQL pg_trgm docs
C10	existing journal lookup tools	Scopus Sources, Web of Science Master Journal List, DOAJ API guides, OpenAlex sources docs

Queries for evaluations of the Committee list and for open-access analyses of Kazakhstani journals returned university news pages and a 2016 policy analysis, not research articles, so none were added. A blog analysis of DOAJ metadata quality (2021) was read but not included because it is not peer-reviewed.

Deviations from the criteria: L8 (Nougat), L16 (OpenAlex) and L20 (fusion functions) are arXiv preprints, L22 is a technical report, L12 is a tool paper, and L4, L9, L11, L13, L19, L23 are older foundational works. L19-L26 were found by targeted search in October 2026 (S13-S15), not by the structured database search, and Kazakh retrieval is covered by two sources only; the search for Kazakh stemming and morphology resources is planned for the thesis (DIP-10). They are flagged in section 3.

Appendix B. Architecture Decision Records

Core track

ADR-001 Retrieval strategy

- **Context:** users search a small multilingual registry with exact, partial, misspelled, transliterated, cross-language and topic queries. The supervisor's document asks for hybrid search with RRF.
- **Options:** FTS only; vector only; hybrid FTS + trigram + vector with fusion (section 5.1).
- **Decision:** hybrid with RRF (k = 60, configurable); keep each single method and a convex-combination variant as baselines and ablations.
- **Consequences:** three retrievers to maintain; evaluation must show value per query type. Revisit if hybrid does not beat the best single method by the HS1 margin.
- **Status:** accepted as design, to be confirmed by measurement.

ADR-002 Search and storage engine

- **Context:** a few hundred rows; needs FTS, fuzzy and vector search; versioned imports.
- **Decision:** PostgreSQL with `pg_trgm` and `pgvector`; exact vector scan first, HNSW only if measured as needed.
- **Consequences:** one database; Kazakh stemming unavailable in standard configs (to verify), so trigram and normalisation carry Kazakh queries.
- **Status:** accepted.

ADR-003 Embedding model

- **Context:** Kazakh, Russian and English text; CPU deployment likely.
- **Decision:** measure CPU latency and memory of BGE M3 and multilingual E5 (small and base) first (task DIP-38, by M1), then evaluate the remaining candidates on the evaluation set and choose the cheaper one if relevance is similar.
- **Consequences:** model choice is delayed until the first evaluation (Jan 2027); embeddings stored with the model name for reproducibility.
- **Status:** open.

ADR-004 Backend framework

- **Context:** Python embedding stack, OpenAPI documentation, supervisor's plan.
- **Decision:** FastAPI with Pydantic models.
- **Consequences:** Node and Express are not used for the API; they remain in the client toolchain and data-integration scripts.
- **Status:** accepted.

ADR-005 Data integration module

- **Context:** the source is an xlsx with an irregular layout (edition of 15 Sep 2026) and later editions may differ; the frontend and data engineer owns the integration.
- **Decision:** Node scripts normalise the source to a versioned JSON contract (`registry-import.schema.json`); the API validates it and stores it.
- **Consequences:** validation rules are defined once (contract); if the source needs Python-only tooling, the parser moves to Python behind the same contract.
- **Status:** proposed.

Appendix C. Abbreviations

Term	Meaning
KOKCHBO MHBO PK	Committee for Quality Assurance in Science and Higher Education, Ministry of Science and Higher Education of Kazakhstan
List 1, 2, 3	levels of the Committee's list of recommended scientific journals (order No. 289 of 3 June 2025)
FTS	full-text search
BM25	classic lexical ranking function used in full-text search
RRF	Reciprocal Rank Fusion, score = sum of 1 / (k + rank)
HNSW	hierarchical navigable small world, approximate nearest-neighbour index
pg_trgm / pgvector	PostgreSQL extensions for trigram similarity and vector search
MRR, nDCG, Recall@k	retrieval metrics (section 9)
ISSN / eISSN	serial number of a journal, print and electronic
FR / NFR	functional / non-functional requirement (section 6)
HS1-HS3	hypotheses (section 1, "Hypotheses")
B0, M1-M4, H1-H4	systems compared in section 9

Appendix D. YouTrack Task List

Project key `DIP`. Owners follow the supervisor's role split. No task is marked done unless the work exists.

D.1 Core track

ID	Title	Owner	Expected outcome	Acceptance criterion	Priority	Depends
DIP-1	Analysis and design report (this document)	both	`docs/REPORT.md`	11 sections present, links resolve	M	
DIP-2	Repository hygiene: README, `.gitignore`, config examples	Nurzhan	files in repo	secret scan clean; README has required fields	M	
DIP-3	Scope questions answered by supervisor	both	answers in section 11 log	open scope questions closed	M	DIP-1
DIP-4	User survey (Google Forms, at least 30 responses) and 3-5 follow-up discussions	both	survey results and notes per discussion; real queries collected	notes use the template; assumptions table updated	M	DIP-1
DIP-5	Repository structure and Docker skeleton (`backend/`, `frontend/`, `integration/`, `deploy/`)	Nurzhan	compose file starts empty services	`docker compose up` runs	M	DIP-2
DIP-6	Import JSON contract `registry-import.schema.json`	both	schema and 3 sample files	sample files validate; invalid sample rejected	M	DIP-3
DIP-7	Database schema and migrations (FR-04, NFR-05)	Nurzhan	migration scripts for registry tables	schema creates; sample kk/ru/en record round-trips	M	DIP-6
DIP-8	Hands-on trial of existing tools on 10 journals; fill section 4	Abilkhaiyr	table of results	each "Not verified" cell in section 4 answered or reasoned	S	
DIP-9	Obtain official list source, record edition, order and format	Abilkhaiyr	file or document copy and note	source and date documented; format decided	M	DIP-3
DIP-10	Extend the literature search for the thesis (more databases, snowballing)	both	updated Appendix A	log lists database, query, hits, selected	S	
DIP-11	Source parser to import JSON (FR-01, FR-02)	Abilkhaiyr	Node script	all rows of the source parsed; invalid rows reported	M	DIP-6, DIP-9
DIP-12	High-fidelity mockups and user flows	Abilkhaiyr	design files	mockups A-D of section 8 reviewed with supervisor	M	DIP-4
DIP-13	Import API with dry-run, versions and diff (FR-01, 03, 12)	Nurzhan	endpoints	two imports with 3 known changes give exact diff	M	DIP-7, DIP-11
DIP-14	Query normaliser, ISSN module, tests (FR-09, NFR-10)	Nurzhan	module with unit tests	each Kazakh letter and ISSN variant test passes	M	DIP-7
DIP-15	Vue skeleton, routing, API client with mock server	Abilkhaiyr	app runs against mock	pages render with mock data	M	DIP-6
DIP-16	FTS retriever (FR-05)	Nurzhan	retriever and test	exact-title queries rank 1 on test data	M	DIP-14
DIP-17	Trigram retriever (FR-06)	Nurzhan	retriever and threshold study	typo test set Recall@5 reported	M	DIP-16
DIP-18	Search page: query box, table, sorting, pagination (FR-14)	Abilkhaiyr	page	meets FR-14 criterion	M	DIP-15
DIP-19	Embedding encoder and vector retriever; model comparison A vs B (FR-07)	Nurzhan	module, comparison table	both models run on dev queries; memory and latency recorded	M	DIP-13
DIP-20	Filters (FR-13)	Abilkhaiyr	filter components	filters combine correctly on test data	M	DIP-18
DIP-21	Fusion: RRF and convex option (FR-08)	Nurzhan	module with config	unit test of config switch passes	M	DIP-16, 17, 19
DIP-22	Search and journal APIs with OpenAPI (FR-10, 11)	Nurzhan	endpoints	contract tests pass	M	DIP-21
DIP-23	Journal card with history (FR-15)	Abilkhaiyr	page	all FR-04 fields visible	M	DIP-18
DIP-24	Query set and judgments, prototype part (FR-19 prerequisite)	both	150 queries with truth	agreement measured	M	DIP-4, DIP-13
DIP-25	Evaluation harness (FR-19)	Nurzhan	script and report	one command writes metrics per method, language, type	M	DIP-22, DIP-24
DIP-26	Status check page and endpoint (FR-16, FR-18)	Abilkhaiyr	page	10 known journals answered correctly	M	DIP-22
DIP-27	Validation, loading, empty and error states (FR-17)	Abilkhaiyr	UI states	all four state tests pass	M	DIP-18
DIP-28	Full query set (at least 400 queries) and final judgments	both	frozen set with hash	split by journal; kappa reported	M	DIP-24
DIP-29	Client and server integration, end-to-end tests	both	e2e suite	status check, search, filters pass against real API	M	DIP-26, DIP-27
DIP-30	Performance tuning, HNSW decision, caching (NFR-02)	Nurzhan	latency report	p95 at or below 300 ms or documented gap	M	DIP-25
DIP-31	CSV export and interface language switch (FR-20, FR-21)	Abilkhaiyr	features	export opens with correct Cyrillic; labels switch	C	DIP-29
DIP-32	Deployment with Docker Compose (NFR-03)	Nurzhan	deployment guide	second machine runs the system	M	DIP-29
DIP-33	Usability test and accessibility check (NFR-06)	Abilkhaiyr	test report	3 users x 3 tasks, keyboard pass, 375 px check	S	DIP-29
DIP-34	Final evaluation run and analysis (HS1-HS3)	both	results chapter	test part used once; confidence intervals	M	DIP-28, DIP-30, DIP-37
DIP-35	Thesis and defence materials	both	thesis, slides	supervisor approved	M	DIP-34
DIP-36	OJS enrichment of ISSN, language and website for the IT/CS subset (section 2.6)	Abilkhaiyr	Node script `integration/enrichment/`	ISSNs of the journals with an OAI endpoint extracted with provenance; offline tests pass	S	DIP-9
DIP-37	HS1 feasibility pilot on the development part (section 9.7)	Nurzhan	pilot report with MRR@10 of M1, M2, M3, H1	headroom over the best single method stated; supervisor informed before any change of HS1 wording	M	DIP-13, DIP-24
DIP-38	CPU latency, start-up time and memory of BGE M3 and multilingual E5 small and base (section 5.3)	Nurzhan	measurement table with hardware and library versions	p50 and p95 of single-query encoding for 100 queries; decision which models enter the comparison	M	DIP-5

Appendix E. API Examples (planned)

Shapes only, with **placeholder data**; the API does not exist yet.

E.1 Core track

`GET /api/search?q=jurnal%20of%20informaton%20tehnologies&level=2&method=hybrid&page=1`

```
{
  "query": "jurnal of informaton tehnologies",
  "method": "hybrid",
  "registry_version": {"version_id": 7, "source_date": "<date>"},
  "total": 2,
  "items": [
    {
      "journal_id": "<uuid>",
      "title": {"en": "Journal A of Information Technologies", "ru": "Журнал А информационных технологий"},
      "issn": "0000-0000",
      "level": "2",
      "status": "active",
      "subjects": ["Informatics"],
      "score": 0.0325,
      "matched_by": ["trigram", "fts"]
    }
  ]
}
```

`GET /api/status?q=0000-0000`

```
{
  "found": true,
  "recommended": true,
  "level": "2",
  "as_of": "<import date>",
  "source": {"order_ref": "<order number and date>", "document": "<name>"},
  "journal_id": "<uuid>",
  "disclaimer": "Reflects the imported list, not an official certificate."
}
```

Not found: `{"found": false, "recommended": false, "as_of": "<import date>", "suggestions": [{"journal_id": "<uuid>", "title": "Journal B ...", "score": 0.021}]}`.

`GET /api/journals/{id}` returns the fields of FR-04 plus `history`: `[{"version_id": 7, "change": "level 3 -> 2"}]`.

`POST /api/admin/import?dry_run=true` with header `X-Admin-Key` and a body that follows `registry-import.schema.json`:

```
{
  "source": {"ref": "<document>", "date": "<edition date>", "order_ref": "<order>"},
  "journals": [
    {"titles": [{"lang": "ru", "title": "..."}, {"lang": "en", "title": "..."}],
     "issn": "0000-0000", "publisher": "...", "level": "2", "subjects": ["..."], "status": "active"}
  ]
}
```

Response: `{"dry_run": true, "added": 3, "removed": 2, "changed": 5, "errors": [{"row": 118, "message": "duplicate ISSN"}]}`. Errors: `401` without key, `422` with the list of schema violations, `404` for an unknown journal, `400` for a query longer than the limit.

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Source ids used in section 3

Id	Reference	Id	Reference
L1	Krause et al., 2021	L10	Saier et al., 2023
L2	Choi et al., 2023	L11	Beltagy et al., 2019
L3	Kong et al., 2023	L12	Lopez, 2009
L4	Lo et al., 2020	L13	Tkaczyk et al., 2015
L5	Nicholson et al., 2021	L14	Visser et al., 2021
L6	Choi et al., 2022	L15	Martín-Martín et al., 2021
L7	Alyafeai et al., 2025	L16	Priem et al., 2022
L8	Blecher et al., 2023	L17	Moskaleva et al., 2018
L9	Cohan et al., 2020	L18	Alimova & Brumshteyn, 2020
		L19	Cormack et al., 2009
		L20	Bruch et al., 2022
		L21	Chen et al., 2024
		L22	Wang et al., 2024
		L23	Reimers & Gurevych, 2020
		L24	Muennighoff et al., 2023
		L25	Yeshpanov et al., 2024 (KazQAD)
		L26	Yeshpanov et al., 2022 (KazNERD)

Web pages and documents that the report cites only as links in sections 2 and 4 (Adilet legal-information system, prg.kz, a-articles.kz, Committee presentation at enic-kazakhstan.edu.kz, Elsevier support for Scopus Sources, CASRAI guides, OpenAlex documentation) are listed with their URLs in the tables where they are used.